

“sees” only one half of the operator. In order for interactions to occur at both half-sites, giving tight binding, it is essential that the structural change in the DNA coincides with the dimer organization of the repressor. Thus, constraints on the conformation of the central base pairs come not only from local backbone interactions with the protein but also from the way in which the dimer interface in the protein fixes the relative position and orientation of the two recognition helices and their respective major groove contacts.

In all complexes studied the protein subunit is anchored across the major groove with extensive contacts along two segments of the sugar-phosphate backbone, one to either side of the groove. Hydrogen bonds between the DNA phosphate groups and peptide backbone NH groups are remarkably prevalent in these contacts (Figure 8.16).

One of these interaction regions involves the loop after the recognition helix, residues 40–44 of the repressor (yellow in Figures 8.14 and 8.15a), where three main-chain NH groups form hydrogen bonds with phosphates 9' and 10'. The side chain of Arg 43 in this loop projects into the minor groove and probably stabilizes its compression, by introducing a positive charge between the phosphate groups on opposite sides of the narrow groove. All residues in this loop, which are outside the helix-turn-helix motif, contribute to the surface complementarity between the protein and the sugar-phosphate surfaces of nucleotides 9' and 10'.

These and other interactions between groups on the protein and the DNA sugar-phosphate backbone stabilize the distorted DNA conformation. They involve a large number of residues that are distributed along most of the polypeptide chain. In addition, other residues that are involved in the subunit interactions in the protein dimer ensure that each subunit is properly poised for binding to its half site in the operator DNA. Thus, the “unit” that is responsible for the differential binding to different operator DNA regions is really an entire binding domain, appropriately dimerized, and nearly all the protein–DNA contacts contribute to this specificity.

Conformational changes of DNA are important for differential binding of repressor and Cro to different operator sites

Repressor and Cro distinguish the different operator sites in two distinct ways. The first involves base pair 4, which is A-T everywhere except in one half-site of OR3 (see Table 8.2). Cro, which binds most tightly to OR3, is indifferent to whether there is A-T or G-C at position 4'; repressor, which binds most tightly to OR1, strongly prefers A-T. The conformation of a synthetic DNA containing the OR3 operator site complexed with the repressor DNA-binding fragment, has provided an important clue to the molecular basis for this discrimination. Binding of the 434 repressor fragment to DNA induces different conformational changes in the halves of OR3. The left half, which contains the consensus sequence 5'-ACAA..., has the same conformation as both halves have in other complexes with the repressor fragment, whereas the right half has the conformation (around phosphate 11) that is found in complexes with Cro.

These results indicate that the repressor fragment is unable to impose upon the nonconsensus sequence on the right-hand side of OR3 the DNA conformation that is required for tight binding between repressor and DNA. It thus appears that the weaker binding of repressor to OR3 is a consequence of the resulting less perfect “fit” between protein and DNA backbone (see Figure 8.13).

The second way repressor and Cro distinguish different operator sites involves the central base pairs, which are the only source of variation among sites other than in OR3. The ability of the protein–DNA contacts to accomplish proper changes in the DNA structure, in particular the overwinding in the central region, can be modulated by the nucleotide sequence of the DNA. The overwinding causes a narrowing of the minor groove in the central

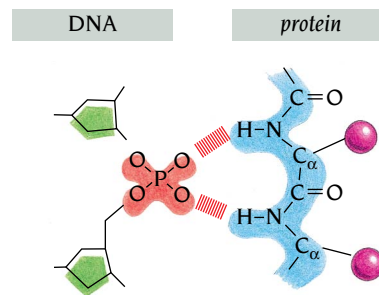


Figure 8.16 Nonspecific protein–DNA interactions are frequently formed by hydrogen bonds (red) between backbone phosphate oxygen atoms of DNA and main-chain NH groups of the protein.

region of the operator. It is generally believed that A-T base pairs can more readily be accommodated in narrow grooves than G-C base pairs. This means that DNA with A-T base pairs in the central region of operator sequences should be able to adopt the conformation necessary for proper DNA-protein interaction more readily than DNA with G-C base pairs in these positions. This is confirmed by mutation experiments using synthetic operator regions; these show that it is possible to change the affinity between repressor and DNA by changing the base pairs in the middle region. Thus, if base pairs 7 and 8 are changed from T-A and A-T to G-C and C-G, the affinity for the repressor fragment is decreased fifteenfold. In other words, the base sequence in this region, by influencing local conformational changes in the DNA, alters the affinity of repressor for DNA. Here we have a case of local DNA structure, rather than direct sequence-specific DNA-protein interactions, modulating repressor binding.

The essence of phage repressor and Cro

The DNA-binding proteins Cro and repressor of the three phages lambda, P22, and 434 all recognize operator regions in DNA through a helix-turn-helix motif. The second α helix of this motif, the recognition helix, binds in the major groove of DNA. These proteins form dimers such that two identical recognition helices, 34 Å apart at the two ends of the elongated dimer, can bind to the ends of the operator regions.

The first two amino acid residues of the recognition helix in these two proteins from phage 434 are glutamines, which form specific hydrogen bonds to the first two base pairs of the operator region. These interactions are identical in all operator regions of phage 434. A few residues on the recognition helix of the helix-turn-helix motif thus form the recognition signal of Cro and repressor proteins for the operator regions by specific interaction with the edge of the base pairs in the major groove of DNA, as predicted by Matthews.

Such sequence-specific interactions do not, however, account for the differential affinity of Cro and repressor for the operator regions. These affinities are instead determined by the ability of the DNA to undergo specific structural changes. At least two factors are important to the ability of the complex to achieve the proper DNA conformation (Figure 8.17). Interactions between the sugar-phosphate backbone of DNA and regions of the protein form a large interaction area of complementary surfaces that stabilizes the structural change in the DNA. These protein regions are not restricted to the

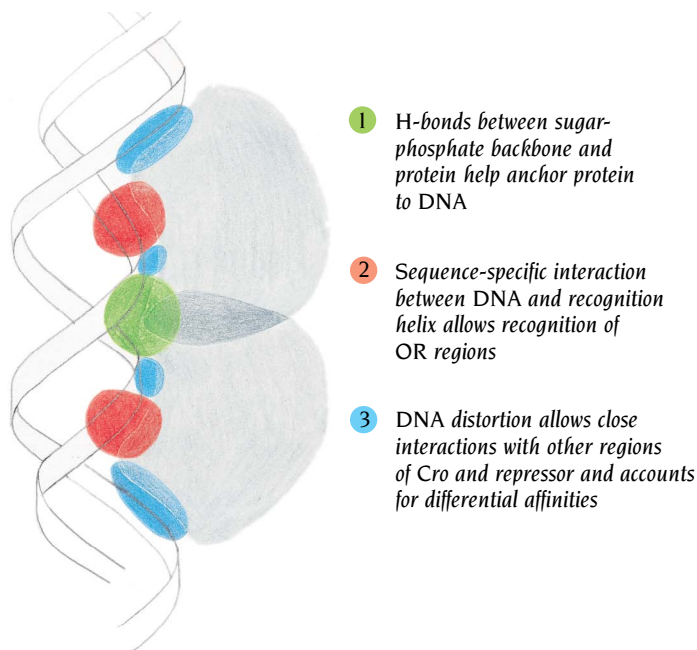


Figure 8.17 Schematic diagram of the main features of the interactions between DNA and the helix-turn-helix motif in DNA-binding proteins.

helix-turn-helix motif but are spread over almost the entire polypeptide chain. The ability of these interactions to accomplish a structural change can be modulated by the sequence of DNA.

DNA binding is regulated by allosteric control

The DNA-binding capacity of most repressors and activators is regulated by small molecules, such as sugars, amino acids, or cyclic AMP, which bind to a distinct site on the protein. This binding causes a conformational change which alters the DNA binding site and therefore regulates its affinity for DNA. As described in Chapter 6, Jacques Monod, at the Pasteur Institute in Paris, introduced the term allosteric binding to emphasize that the small ligands, known as **allosteric effectors**, and the sites at which they bind are sterically quite distinct from the functional binding sites. We will discuss one example of allosteric control of DNA binding—the tryptophan (*trp*) repressor from *Escherichia coli*.

The *trp* repressor forms a helix-turn-helix motif

The ***trp* repressor** controls the operon for the synthesis of L-tryptophan in *Escherichia coli* by a simple **negative feedback loop**. In the absence of L-tryptophan, the repressor is inactive, the operon is switched on and the enzymes which synthesize L-tryptophan are produced. As the concentration of L-tryptophan increases, it binds to the repressor and converts it to an active form so that it can bind to the operator region and switch off the gene.

The structure of the *trp* repressor has been determined both with and without bound tryptophan to 1.8 Å resolution by the group of Paul Sigler, then at the University of Chicago. The *trp* repressor is a dimer, like the other proteins we have discussed in this chapter. The polypeptide chain has 107 amino acids and is folded into six α helices (Figure 8.18). It therefore has an α -type structure like the lambda repressor, but the arrangement of the helices in the subunit is quite different. The six helices do not pack to form a regular structure with a hydrophobic core, and it is doubtful if monomers with the arrangement of helices shown in Figure 8.18 would be stable alone. Stability is conferred by dimerization because the two subunits fit together to give a functional molecule that, in contrast to its subunits, has a compact globular form (Figure 8.19).

The helices at the N-terminal regions of the two polypeptide chains are intertwined and make extensive contacts in the central part of the molecule to form a stable core. This core supports two “heads”, each comprising the last three helices from one polypeptide chain. Alpha helix 3 in the middle of the subunit chain is quite long and forms the main link between the core and the head.

Structural studies of a repressor–DNA complex have shown that helices 4 and 5 form a helix-turn-helix motif and that side chains from the recognition helix 5 form water-mediated interactions with bases in the major groove.

A conformational change operates a functional switch

What is the molecular mechanism by which the binding of tryptophan—the allosteric effector—allows the *trp* repressor to bind DNA? The structures of the *trp* repressor with and without bound tryptophan provided a simple answer. In the absence of tryptophan the heads, and in particular the recognition helices, are tilted inward toward the core (Figure 8.20a). This makes the distance between the two DNA binding sites of the repressor molecule too short by 5–6 Å to allow them to fit into the major groove. Specific DNA binding is not possible, and the repressor is inactive. Tryptophan binding causes a conformational change in the repressor that alters the orientations of the recognition helices. The heads tilt outward and two identical cavities are formed between the heads and the core (Figure 8.20b). Two tryptophan molecules

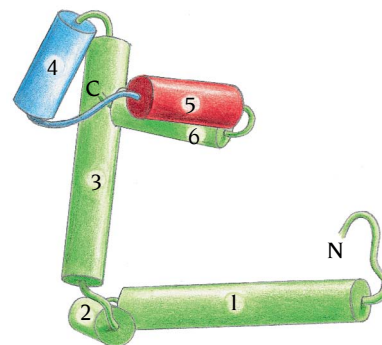
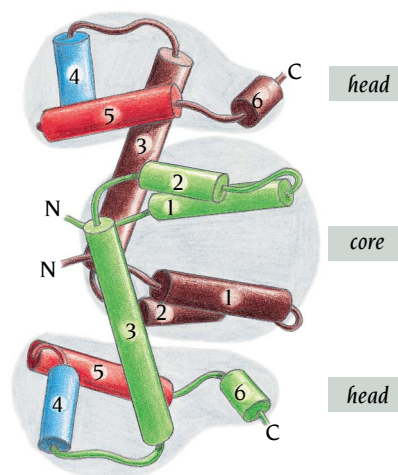
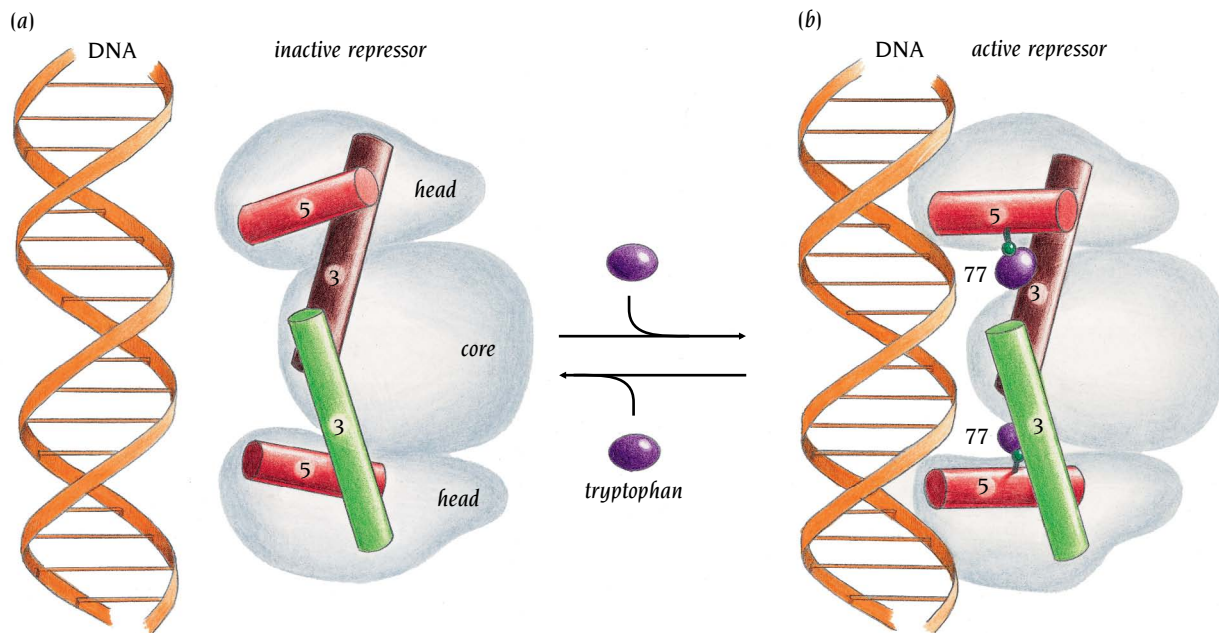


Figure 8.18 The subunit of the *trp* repressor. The subunit contains 107 amino acid residues that are folded into six α helices. Helices 4 (blue) and 5 (red) form the DNA-binding helix-turn-helix motif. (Adapted from R.W. Schevitz et al., *Nature* 317: 782–786, 1985.)

Figure 8.19 The α helices of the N-terminal region of the *trp* repressor are involved in subunit interactions and form a stable core in the middle of the dimer. Alpha helices 4–6, which include the helix-turn-helix motif, form two “head” regions at the two ends of the molecule. Alpha helix 3 connects the core to the head in both subunits. (Adapted from R.W. Schevitz et al., *Nature* 317: 782–786, 1985.)





bind to the dimeric repressor in these cavities; they are wedged between the long connecting α helices (3) and the recognition helices of the heads. When tryptophan is bound in these cavities, the recognition helices are properly poised 34 Å apart for binding DNA in the major groove, as they are in the 434 repressor.

The elegant genetic studies by the group of Charles Yanofsky at Stanford University, conducted before the crystal structure was known, confirm this mechanism. The side chain of Ala 77, which is in the loop region of the helix-turn-helix motif, faces the cavity where tryptophan binds. When this side chain is replaced by the bulkier side chain of Val, the mutant repressor does not require tryptophan to be able to bind specifically to the operator DNA. The presence of a bulkier valine side chain at position 77 maintains the heads in an active conformation even in the absence of bound tryptophan. The crystal structure of this mutant repressor, in the absence of tryptophan, is basically the same as that of the wild-type repressor with tryptophan. This is an excellent example of how ligand-induced conformational changes can be mimicked by amino acid substitutions in the protein.

Figure 8.20 Schematic diagrams of docking the *trp* repressor to DNA in its inactive (a) and active (b) forms. When L-tryptophan, which is a corepressor, binds to the repressor, the “heads” change their positions relative to the core to produce the active form of the repressor, which binds to DNA. The structures of DNA and the *trp* repressor are outlined. The positions of the DNA recognition helices (5) and the helices (3) that connect the core with the heads are indicated. The approximate position of the side chain of residue 77 is marked as a purple ball (see text for the significance of this residue). (Adapted from R.-G. Zhang et al., *Nature* 327: 591–597, 1987.)

Lac repressor binds to both the major and the minor grooves inducing a sharp bend in the DNA

More than 30 years ago Jacob and Monod introduced the *Escherichia coli* *lac* operon as a model for gene regulation. The *lac* repressor molecule functions as a switch, regulated by inducer molecules, which controls the synthesis of enzymes necessary for *E. coli* to use lactose as an energy source. In the absence of lactose the repressor binds tightly to the operator DNA preventing the synthesis of these enzymes. Conversely when lactose is present, the repressor dissociates from the operator, allowing transcription of the operon.

The *lac* repressor monomer, a chain of 360 amino acids, associates into a functionally active homotetramer. It is the classic member of a large family of bacterial repressors with homologous amino acid sequences. PurR, which functions as the master regulator of purine biosynthesis, is another member of this family. In contrast to the *lac* repressor, the functional state of PurR is a dimer. The crystal structures of these two members of the Lac I family, in their complexes with DNA fragments, are known. The structure of the tetrameric *lac* repressor–DNA complex was determined by the group of Mitchell Lewis, University of Pennsylvania, Philadelphia, and the dimeric PurR–DNA complex by the group of Richard Brennan, Oregon Health Sciences University, Portland.

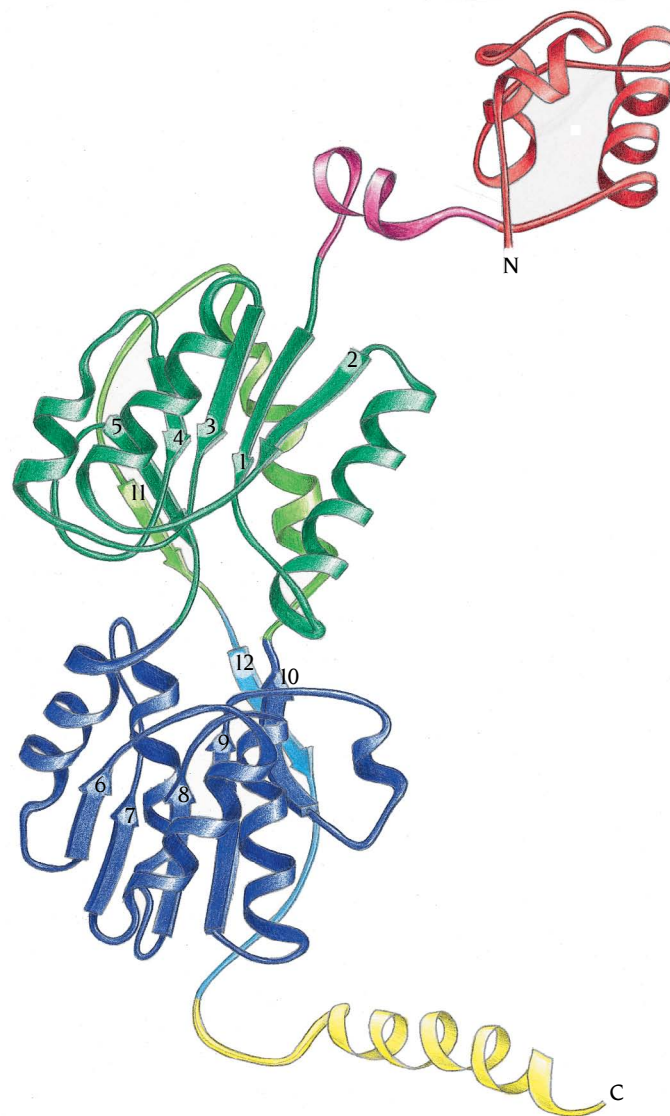


Figure 8.21 Richardson-type diagram of the structure of one subunit of the lac repressor. The polypeptide chain is arranged in four domains, an amino terminal DNA-binding domain (red) with a helix-turn-helix motif, a hinge helix (purple), a large core domain which has two subdomains (green and blue) and a C-terminal α helix. (Adapted from M. Lewis et al., *Science* 271: 1247–1254, 1996.)

The polypeptide chain of the lac repressor subunit is arranged in four domains (Figure 8.21): an N-terminal DNA-binding domain with a helix-turn-helix motif, a hinge helix which binds to the minor groove of DNA, a large core domain which binds the corepressor and has a structure very similar to the periplasmic arabinose-binding protein described in Chapter 4, and finally a C-terminal α helix which is involved in tetramerization. This α helix is absent in the PurR subunit structure; otherwise their structures are very similar.

The tetrameric structure of the lac repressor has a quite unusual V-shape (Figure 8.22). Each arm of the V-shaped molecule is a tight dimer, which is very similar in structure to the PurR dimer and which has the two N-terminal DNA binding domains close together at the tip of the arm. The two dimers of the lac repressor are held together at the other end by the four carboxy-terminal α helices, which form a four-helix bundle.

Each dimer arm binds a separate copy of a palindromic DNA sequence. The recognition helices of the two subunits insert into successive major grooves in a way similar to the binding of bacteriophage Cro and repressor molecules described above. In addition, the two hinge helices of the dimer, which are close together, interact with the minor groove between the two major-groove binding sites of the helix-turn-helix motifs (Figure 8.23). Both the recognition helix and the hinge helix form sequence-specific interactions between their amino acid side chains and the nucleotide bases of the DNA.

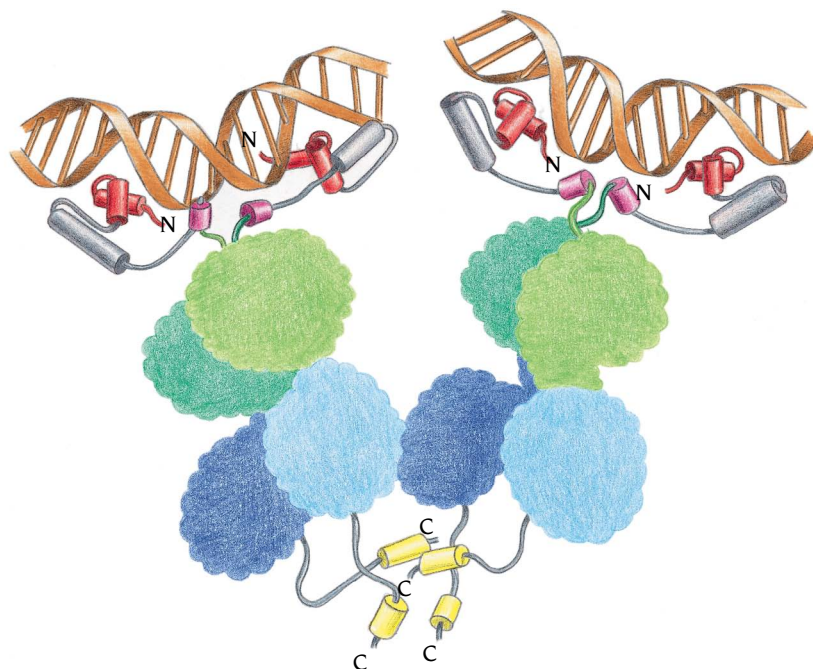


Figure 8.22 The lac repressor molecule is a V-shaped tetramer in which each arm is a dimer containing a DNA-binding site. The helix-turn-helix motifs (red) of each dimer bind in two successive major grooves and the hinge helices (purple) bind adjacent to each other in the minor groove between the two major groove binding sites. The four subunits of the tetramer are held together by the four C-terminal helices (yellow) which form a four helix bundle. The bound DNA fragments are bent. (Adapted from M. Lewis et al., *Science* 271: 1247–1254, 1996.)

This is an example of minor groove base recognition in the context of major groove recognition. Two bulky leucine residues from the hinge helices force the minor groove to open up and become wide and shallow. As a result the DNA is distorted from its B-form and sharply bent away from the protein (see Figure 8.22). Due to this bending, the distance between the two recognition helices in the major groove and their relative orientation are different from those in bacteriophage Cro and repressor complexes with DNA. The wild-type operator regions for the lac repressor are pseudo-palindromic with a central base pair between the pseudo-symmetric halves whereas the DNA in the crystals is strictly palindromic and lacks the central base pair. This difference would not affect binding of the recognition helices to DNA, but details of the DNA bending and binding of the hinge helices in the central part of the DNA could be somewhat different in the crystals compared with the situation *in vivo*. However, such differences are expected to be minor since the dimeric PurR repressor also bends DNA and essentially looks like one arm of the tetrameric lac repressor. The DNA molecule in the crystals of the PurR–DNA complex has the same number of base pairs as the PurR operator regions. The two arms of the V-shaped tetramer of the lac repressor each bind their DNA fragments in a similar way. Since the binding sites are far away from each other in the tetramer at the tips of the two arms of the V there is no interaction between the two DNA fragments (see Figure 8.22).

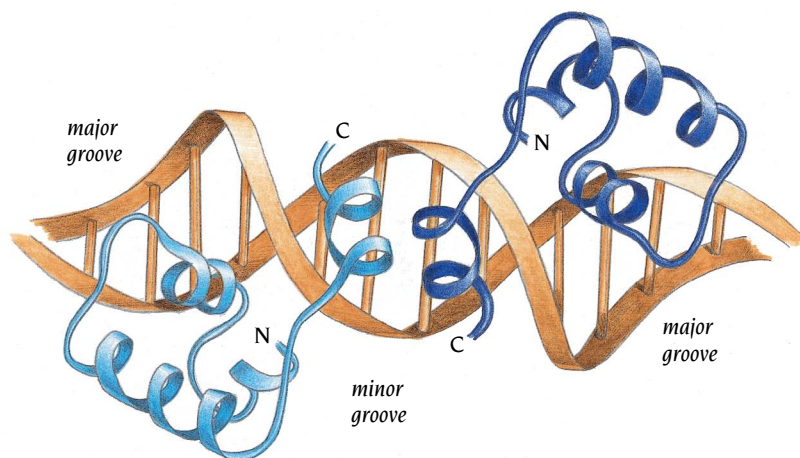


Figure 8.23 The helix-turn-helix motifs of the subunits of both the PurR and the lac repressor subunits bind to the major groove of DNA with the N-terminus of the second helix, the recognition helix, pointing into the groove. The two hinge helices of each arm of the V-shaped tetramer bind adjacent to each other in the minor groove of DNA, which is wide and shallow due to distortion of the B-DNA structure. (Adapted from M.A. Schumacher et al., *Science* 266: 763–770, 1994.)

CAP-induced DNA bending could activate transcription

CAP, catabolite gene activating protein, is a DNA-binding protein that assists RNA polymerase in binding effectively to certain *Escherichia coli* promoters. In other words, it is a **positive control protein** that promotes more frequent initiation of RNA synthesis. Alone, CAP is a nonspecific DNA-binding protein but it is converted by binding cyclic AMP to a form that binds strongly to specific operator regions.

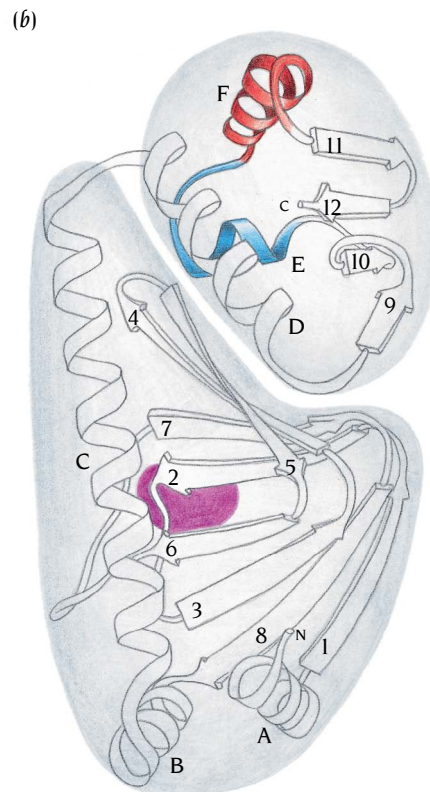
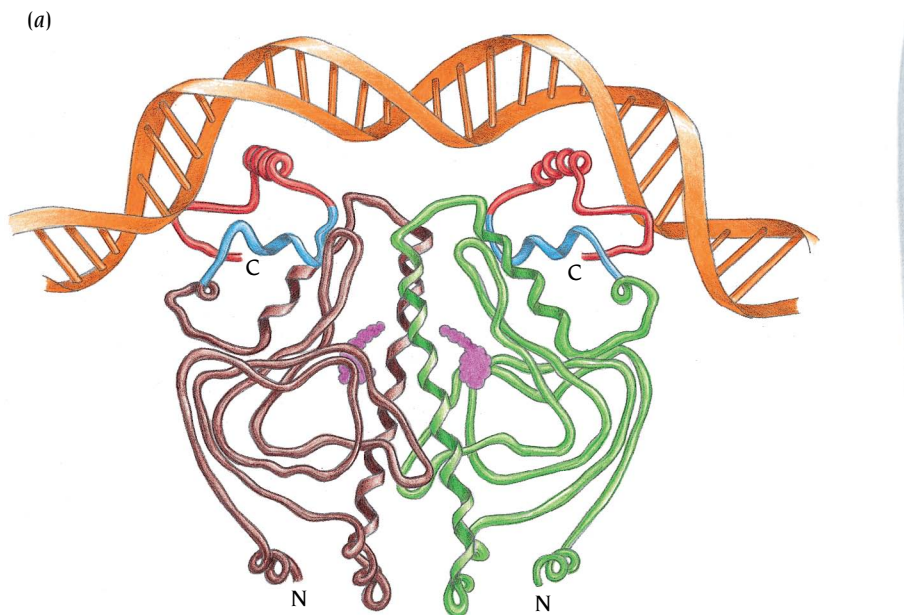
CAP controls a number of operons, all of which are involved in the breakdown of sugar molecules and one of which is the lac operon. When the level of the breakdown products of lactose is low, the concentration of cyclic AMP in the cell increases and CAP is switched on, binds to its specific operators, and increases the rate of transcription of adjacent operons.

Many biochemical and biophysical studies of CAP-DNA complexes in solution have demonstrated that CAP induces a sharp bend in DNA upon binding. This was confirmed when the group of Thomas Steitz at Yale University determined the crystal structure of cyclic AMP-DNA complex to 3 Å resolution. The CAP molecule comprises two identical polypeptide chains of 209 amino acid residues (Figure 8.24). Each chain is folded into two domains that have separate functions (Figure 8.24b). The larger N-terminal domain binds the allosteric effector molecule, cyclic AMP, and provides all the subunit interactions that form the dimer. The C-terminal domain contains the helix-turn-helix motif that binds DNA.

The CAP dimer interacts directly with 27 of the 30 base pairs of the DNA fragment present in the crystal. The amino ends of the recognition helices (red in Figure 8.24) of the helix-turn-helix motifs are bound in the major groove of DNA. These interactions provide sequence specificity between protein side chains and the exposed edges of base pairs. In addition 22 of the phosphate groups of DNA interact directly with protein side chains or main-chain amide groups providing a large number of contacts.

The central 10 base pairs of the palindromic DNA molecule have a regular B-DNA structure. Between base pairs 5 and 6 in each half of the fragment (base pairs are counted from the center) there is a 40° kink which causes these base pairs to be unstacked (Figure 8.24a). After this localized kink the two end regions have an essentially B-DNA structure. The kink occurs at a TG step in the sequence GTG. These TG steps at positions 5 and 6 are highly conserved in both halves of different CAP-binding sites, presumably in part because they facilitate kinking.

Figure 8.24 (a) Structure of the CAP-cyclic AMP-DNA complex. The two protein chains of the dimeric CAP molecule are colored green and brown. The helix-turn-helix motif is blue and red with the recognition helix and the C-terminus red. The DNA which is kinked where the recognition helices bind is orange and the cyclic AMP molecules are purple. (b) Schematic diagram of the structure of one subunit of CAP. The β strands (1–12) and α helices (A–F) are labeled from the N-terminus. The DNA-binding helix-turn-helix motif is colored blue and red with the recognition helix red. The binding site for cyclic AMP is purple. [(a) Adapted from S.C. Schultz et al., *Science* 253: 1001–1007, 1991. (b) Adapted from D. McKay and T. Steitz, *Nature* 290: 744–749, 1981.]



What contributes to the energy of bending in this complex? The central 10 base pairs are anchored to the protein by hydrogen bonds to the six DNA phosphates that face the protein in this region. The surface of the protein is complementary to B-DNA so that no bending of DNA is required to form these interactions. In contrast, interactions between CAP and the 10 base pairs at each end of DNA require that the DNA be bent. These base pairs interact extensively with the protein to provide in each half of the molecule five ionic interactions and four hydrogen bonds to DNA phosphates. These interactions could not occur with straight DNA and therefore can contribute to binding only if the DNA is bent. Thus, the energy required to bend the DNA in this complex is largely supplied by extensive hydrogen bonding and ionic interactions with the DNA phosphates of the two flanking segments.

Steitz has suggested that DNA bending by CAP could contribute to activation of transcription by looping the DNA around CAP to provide for contacts between RNA polymerase and DNA upstream of the CAP-binding site. Such a model could explain how CAP can activate transcription from a variety of distances from the RNA polymerase-binding site since the size of the loop could vary.

Conclusion

Many proteins that switch off or on gene expression in bacteria are dimeric molecules, and the DNA sequences that they specifically recognize are palindromic at their ends. The twofold symmetry of the protein is therefore matched by twofold symmetry at the ends of the recognition sequence.

The monomeric subunits have a helix-turn-helix motif that functions as the specific DNA-binding region. The second helix of this motif is the recognition helix, and its side chains form hydrogen bonds and hydrophobic contacts with nucleotide bases at the bottom of the major groove in B-DNA. These interactions form the recognition signal of Cro and repressor proteins for the operator regions. In the dimeric protein molecules recognition helices in each of the two helix-turn-helix motifs, at opposite ends of the elongated molecule, are separated by 34 Å, corresponding to one turn of B-DNA. When one recognition helix binds to the DNA major groove, the second, 34 Å away, is positioned to bind into the major groove one helical turn along the DNA.

Binding of 434 Cro or repressor to operator sites imposes specific structural changes on the DNA. The changes imposed by the two proteins are overall very similar, but they differ in one segment of the DNA backbone, that is important for the key specificity difference between Cro and repressor (their relative affinities for OR1 and OR3). The conformation of bound DNA is stabilized by interactions between its sugar-phosphate backbone and regions of the protein, which form a large interaction area with complementary surfaces. The ability of these interactions to accomplish a structural change can be modulated by the actual sequence of the DNA, accounting for affinity differences among the various sites.

Specific recognition of DNA targets by the helix-turn-helix motif involves not only interactions between side chains of the recognition helix and bases in the major groove of DNA, but is to a large extent governed by interactions within complementary surfaces between the protein and the DNA. These interactions frequently involve hydrogen bonds from protein main-chain atoms to the DNA backbone in both the major and minor grooves, and are dependent on sequence-specific deformability of the target DNA. Water-mediated interactions often occur even between bases in the major groove and side chains of the recognition helix.

Some of the procaryotic DNA-binding proteins are activated by the binding of an allosteric effector molecule. This event changes the conformation of the dimeric protein, causing the helix-turn-helix motifs to move so that they are 34 Å apart and able to bind to the major groove. The dimeric repressor for purine biosynthesis, PurR, induces a sharp bend in DNA upon binding caused by insertion of α helices in the minor groove between the two

helix-turn-helix motif binding sites in the major groove. Each dimer of the V-shaped tetrameric lac repressor, which has sequence homology to PurR, binds independently to DNA fragments in a similar way to the PurR repressor dimer. Such induced bending of DNA may allow the lac repressor molecule to bind to two operator sites which are distant from each other in the operon.

Binding of the catabolite gene activator protein, CAP, to DNA bends the DNA by producing two localized kinks each of 40°. Such DNA bending could activate transcription by facilitating contacts between RNA polymerase and DNA upstream from the CAP binding site. Sequence specific binding of CAP derives both from sequence-dependent distortions of the DNA helix and from direct sequence-specific interactions between protein side chains and exposed edges of base pairs in the major groove.

Genetics and model building provide a framework for the interpretation of structural data, but they cannot reliably be used to predict the details of the stereochemistry of specific interactions within or between molecules.

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DNA Recognition by Eucaryotic Transcription Factors

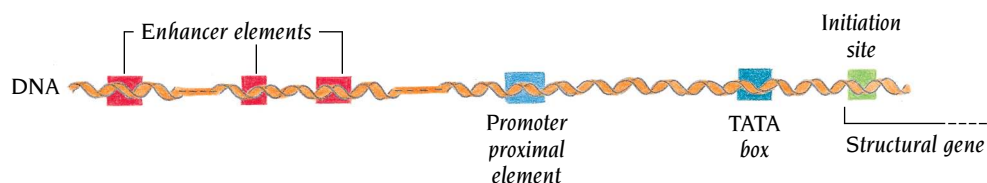
9

The regulation of transcription in eucaryotes is in general much more complex and currently less well understood than the rather simple switch mechanisms that regulate procaryotic gene expression, examples of which we discussed in Chapter 8. There are three classes of RNA polymerases in eucaryotes. Two of these polymerases, RNA polymerase I and RNA polymerase III, transcribe the genes encoding ribosomal RNAs and transfer RNAs, respectively; genes that code for the messenger RNAs of proteins are transcribed by RNA polymerase II, and it is these genes that will be our principal focus. Complex sets of regulatory elements control the initiation of transcription of these structural genes. Distal to the RNA polymerase II initiation site there are different combinations of specific DNA sequences, each of which is recognized by a corresponding site-specific DNA-binding protein. These proteins are called **transcription factors**, and each combination of DNA sequence and cognate transcription factor constitutes a **control module**. The essence of transcriptional regulation in eucaryotes is to use different combinations of a large set of control modules to regulate the expression of each gene. Given the large number of modules that is now being discovered in the human genome, the number of possible combinations is almost unlimited.

The DNA part of each control module can be divided into three main regions, the **core** or **basal promoter elements**, the **promoter proximal elements** and the **distal enhancer elements** (Figure 9.1). The best characterized core promoter element is the **TATA box**, a DNA sequence that is rich in A-T base pairs and located 25 base pairs upstream of the transcription start site. The TATA box is recognized by one of the basal transcription factors, the **TATA box-binding protein, TBP**, which is part of a multisubunit complex called **TFIID**. This complex in combination with RNA polymerase II and other basal transcription factors such as **TFIIA** and **TFIIB** form a preinitiation complex for transcription.

The promoter proximal elements are usually 100 to 200 base pairs long and relatively close to the site of initiation of transcription. Within each of these elements there are DNA sequences specifically recognized by several different transcription factors which either interact directly with the preinitiation complex or indirectly through other proteins.

Figure 9.1 The transcriptional elements of a eucaryotic structural gene extend over a large region of DNA. The regulatory sequences can be divided into three main regions: (1) the basal promoter elements such as the TATA box, (2) the promoter proximal elements close to the initiation site, and (3) distal enhancer elements far from the initiation site.



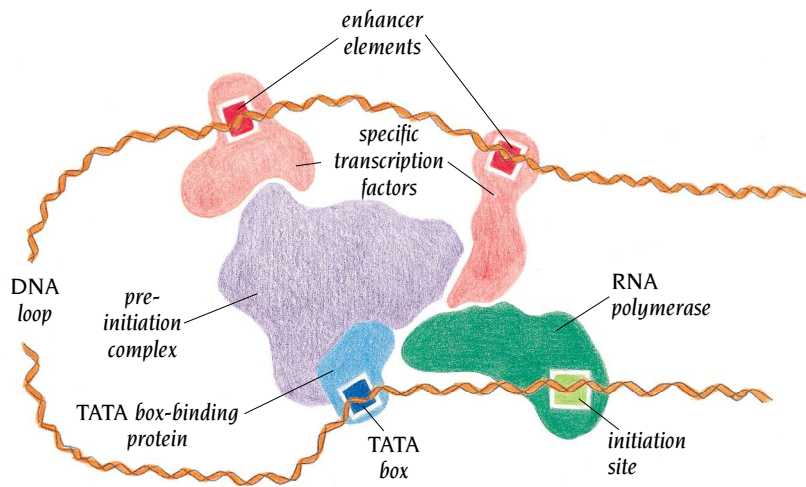


Figure 9.2 Schematic model for transcriptional activation. The TATA box-binding protein, which bends the DNA upon binding to the TATA box, binds to RNA polymerase and a number of associated proteins to form the preinitiation complex. This complex interacts with different specific transcription factors that bind to promoter proximal elements and enhancer elements.

Enhancer elements by contrast are short DNA sequences that occur further upstream or downstream from the initiator site than the promoter proximal elements. Enhancers contain specific sequences recognized by cognate transcription factors. The remarkable feature of enhancers is their distance from the promoters they control. They are often a few thousand base pairs from the promoter but may be 20,000 base pairs or more distant. This means that in eucaryotes transcription is regulated by a series of DNA sequences, each specifically recognized by a DNA-binding protein, dispersed over very long stretches of DNA (Figure 9.2). Efficient gene expression, therefore, can take place only in cells in which the appropriate DNA-binding proteins are present. Every cell of an organism has the same DNA control sequences, but not the complete or the same set of DNA-binding proteins. The regulation of transcription, including cell-specific gene expression, depends on the complement of transcription factors present at any one time in each cell.

Transcription factors can be divided into two broad classes. The general factors of the preinitiation complex are required for the expression of all structural genes transcribed by RNA polymerase II and are, therefore, ubiquitous. Each specific transcription factor that binds either to promoter proximal DNA sequences or to distal enhancer elements is present in only one or a few types of cells, or in cells under particular physiological or environmental conditions. The presence or absence of such transcription factors and their activation or inactivation therefore determine most of the functions of eucaryotic cells, from the changes they undergo during development and differentiation in a multicellular organism to their responses to environmental changes, such as heat shock.

Transcription is activated by protein-protein interactions

The general transcription factor TFIID is believed to be the key link between specific transcription factors and the general preinitiation complex. However, the purification and molecular characterization of TFIID from higher eucaryotes have been hampered by its instability and heterogeneity. All preparations of TFIID contain the TATA box-binding protein in combination with a variety of different proteins called **TBP-associated factors, TAFs**. When the preinitiation complex has been assembled, strand separation of the DNA duplex occurs at the transcription start site, and RNA polymerase II is released from the promoter to initiate transcription. However, TFIID can remain bound to the core promoter and support rapid reinitiation of transcription by recruiting another molecule of RNA polymerase.

The polypeptide chains of the specific transcription factors usually have two different functions: one is to bind to a specific DNA sequence and another is to activate transcription. These two functions are often

embodied in separate domains of the protein. How the activating domains function to increase the initiation of transcription is not well understood, but in higher eucaryotes it is believed that they often act through interactions with TAFs to enhance the binding of RNA polymerase to the promoter by contributing to the formation of a stable initiation complex at the TATA box (see Figure 9.2). When the specific transcription factor is part of a DNA control module many hundreds of base pairs away from the promoter site, the intervening DNA is thought to loop out to enable the activating region of the protein to participate in the formation of the transcriptional initiation complex.

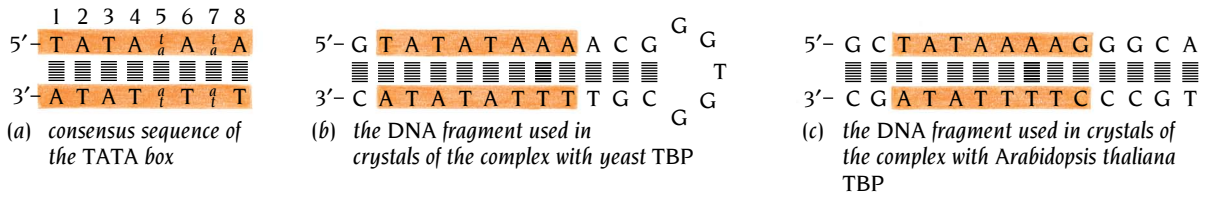
A complete transcriptional initiation complex comprising many different protein molecules as well as a large DNA molecule is very difficult to reconstitute *in vitro* and has not yet been crystallized. We therefore know little about the interactions in such complexes of the transcription factors with each other, with RNA polymerase, or with TAFs. Biochemical evidence from cooperative binding studies indicates that there are defined spatial relationships between the different proteins, but as yet we do not understand them at atomic resolution. However, we have made considerable progress toward understanding the structural basis for the recognition of target DNA sequences by specific transcription factors. The DNA-binding domains of the transcription factors are usually short polypeptide chains of around 100 amino acids and they bind to short DNA regions of less than 20 nucleotides. These protein and DNA molecules are easily produced by recombinant DNA technology and chemical synthesis, respectively. The structures of a large number of such protein:DNA complexes have recently been determined by NMR and x-ray techniques. Although the protein molecules used in these studies are only fragments of complete transcription factors, we know from biochemical and genetic studies that their DNA-binding properties in general are the same as those of the intact transcription factors.

Genes that encode specific transcription factors are now being cloned and sequenced at a rapid pace. By comparing the deduced amino acid sequences of these proteins it has become apparent that their DNA-binding regions are built up from a very limited number of structural motifs. Both x-ray and NMR studies have given detailed structural information on complexes of DNA with members of the four most important families of specific transcription factors: those with helix-turn-helix motifs as in procaryotes, and the leucine zipper, helix-loop-helix and zinc-containing motifs that we shall discuss in Chapter 10. About 80% of the known transcription factors have DNA-binding domains that belong to one or another of these four families, the first of which we shall discuss later in this chapter, while the next chapter is devoted to the other three classes.

The TATA box-binding protein is ubiquitous

TBP, the TATA box-binding protein, was first isolated and purified from yeast in 1988 and found to be composed of a single polypeptide chain with a molecular weight of 27 kDa. Yeast TBP is capable of binding specifically to the TATA box of many promoters. Isolation of the corresponding cDNA clone led quickly to the identification of genes encoding homologous proteins from a wide variety of species including those of plants, insects and mammals. Comparison of the deduced amino acid sequences of TBPs reveals that they are composed of a highly conserved C-terminal domain of 180 amino acids and an N-terminal domain that varies in length and shows little or no sequence conservation among different classes of organisms.

TBP mutants lacking the N-terminal region are fully functional in promoter binding and stimulation of basal transcription and therefore these two functions must be provided by the C-terminal domain. Furthermore, the C-terminal domain of yeast TBP contains all the functions essential for normal yeast cell growth and for responses to specific transcriptional activators with a net negative charge. This C-terminal domain contains two homologous



repeats of 88 amino acids each and mutation studies have indicated that these repeats bind to the TATA consensus sequence (Figure 9.3a) with a dissociation constant in the nanomolar range.

The three-dimensional structures of TBP–TATA box complexes are known

The three-dimensional structures of two complexes between DNA fragments containing a TATA box and the C-terminal domain of TBP were determined in 1993. The group of Paul Sigler at Yale University, worked with yeast TBP whilst the group of Stephen Burley at Rockefeller University, used TBP from the plant *Arabidopsis thaliana*. Both groups used recombinant proteins that were produced in *Escherichia coli*. Sigler's group produced complete yeast TBP then enzymatically cleaved off the N-terminal domain. The C-terminal domain was cocrystallized with a 29 nucleotide hairpin DNA sequence bearing one of the TATA boxes present in the yeast genome (Figure 9.3b). The plant protein studied by Burley had the N-terminal region of 20 amino acids preceding the DNA-binding C-terminal domain, and was cocrystallized with a 14 base pair construction containing the TATA element of the adenovirus late promoter (see Figure 9.3c). Much of the short N-terminal segment was not visible in the crystal structure, presumably because it is disordered. The two structures are surprisingly similar bearing in mind that one protein is from yeast, the other from a plant, and that the two TATA boxes are from such diverse sources as yeast and an animal virus. The results are therefore in all probability valid for all TBP–TATA box interactions, and provide a framework for detailed functional studies of this fundamental part of gene regulation in all species.

A β sheet in TBP forms the DNA-binding site

The two homologous repeats, each of 88 amino acids, at both ends of the TBP DNA-binding domain form two structurally very similar motifs. The two motifs each comprise an antiparallel β sheet of five strands and two helices (Figure 9.4). These two motifs are joined together by a short loop to make a 10-stranded β sheet which forms a saddle-shaped molecule. The loops that connect β strands 2 and 3 of each motif can be visualized as the stirrups of this molecular saddle. The underside of the saddle forms a concave surface built up by the central eight strands of the β sheet (see Figure 9.4a). Side chains from this side of the β sheet, as well as residues from the stirrups, form the DNA-binding site. No α helices are involved in the interaction area, in contrast to the situation in most other eucaryotic transcription factors (see below).

The side of the β sheet that faces away from DNA is covered by two long α helices. One of these helices contains a number of basic residues from the middle segment of the polypeptide chain while the second helix is formed by the C-terminal residues. Residues from these two helices and from the short loop that joins the two motifs (red in Figure 9.4) are likely candidates for interactions with other subunits of the TFIID complex, and with specific transcription factors.

Figure 9.3 Comparison of the consensus nucleotide sequence of the TATA box (a) and the sequences of the DNA fragments used in the crystal structure determinations of the TATA box-binding proteins from yeast (b) and the plant *Arabidopsis thaliana* (c).

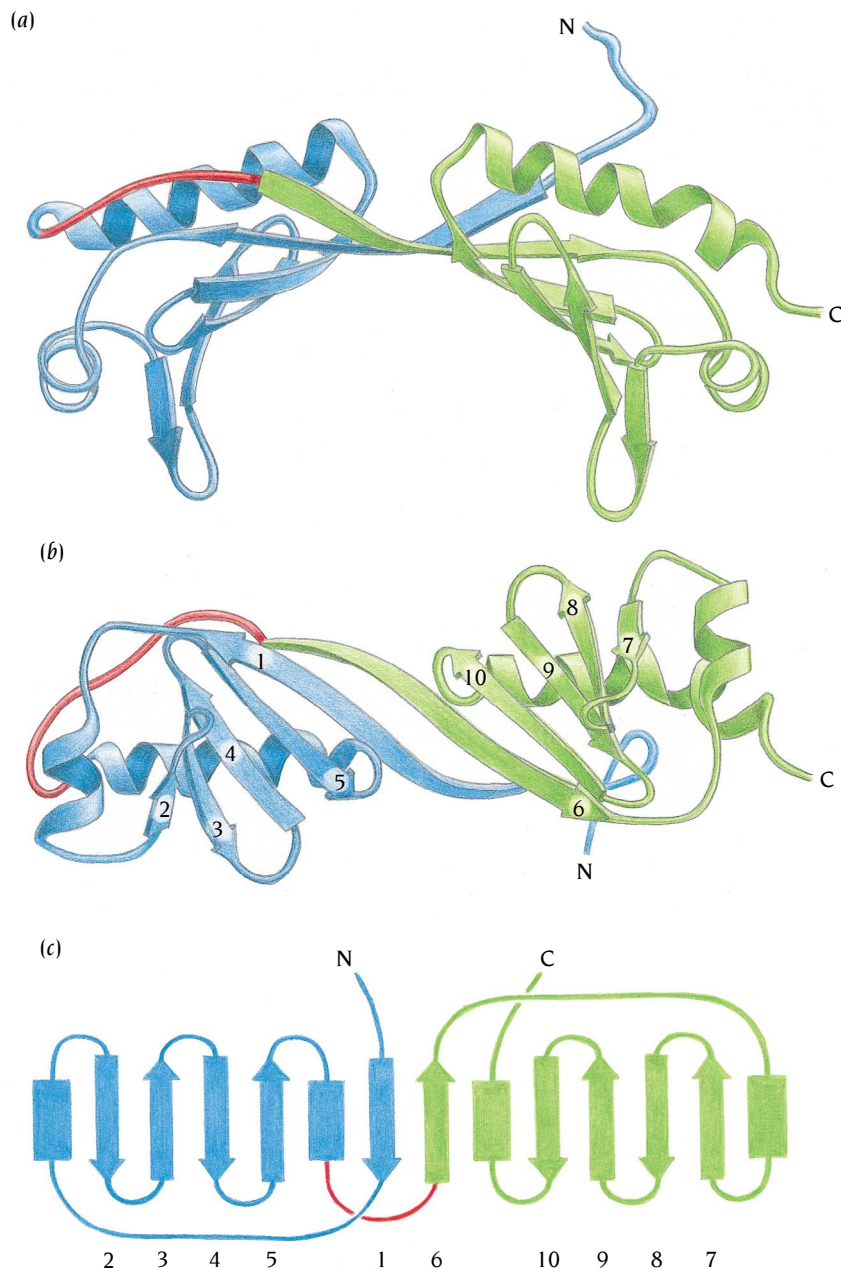


Figure 9.4 Schematic diagrams of the structure of the TATA box-binding protein. (a and b) Richardson-type diagrams illustrating the structural similarity of the two domains (blue and green), which together form a saddle-shaped molecule with stirrups, with the saddle formed from an eight-stranded β sheet. A loop region connecting the two motifs is shown in red. The view in (b) is rotated 90° from that in (a). (c) Topological diagram of the secondary structure elements along the polypeptide chain. (Adapted from D.B. Nikolov et al., *Nature* 360: 40–46, 1992.)

TBP binds in the minor groove and induces large structural changes in DNA

In the two complexes studied by x-ray crystallography, the interactions between TBP and the DNA, as well as the deformation of the B-DNA structure, are very similar, and we will illustrate some of these details for the yeast structure. Minor details of the two complexes vary due to differences in some of the side chains and nucleotides that are present in the interaction areas.

All eight base pairs of the TATA box are in contact with the protein and their structure deviates greatly from the canonical B-form DNA (Figure 9.5). When looking at the TBP structure alone, in the orientation shown in Figure 9.4a, one could imagine that the saddle would straddle normal B-DNA with the helical axis of the DNA perpendicular to a line connecting the two stirrups (perpendicular to the plane of the paper). But this is not the case. Instead, the DNA is sharply bent in the TATA box region so that the local helical axis is almost parallel to the line from stirrup to stirrup. Remarkably, the normal B-DNA structure abruptly returns outside the TATA box. The helical axes of the DNA at each end of the TATA box form an angle of about 100° to

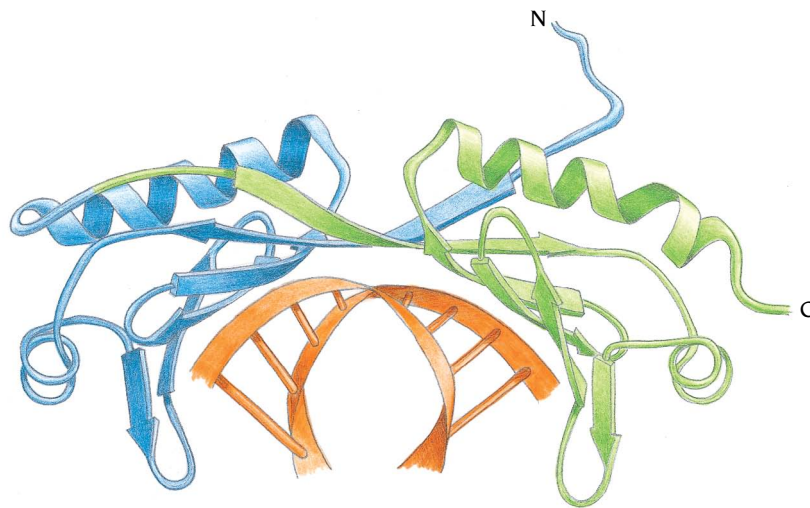


Figure 9.5 Schematic diagram illustrating the structure of the complex between TBP and a DNA fragment containing the TATA box. Both the stirrups and the underside of the saddle are in contact with the DNA. (Adapted from Y. Kim et al., *Nature* 365: 514–520, 1993.)

each other, instead of the expected 180° if the DNA was not bent (Figure 9.6). Thus the deformation is restricted to and persists over the entire TATA box, producing severe bends that radically alter the direction taken by the flanking B-DNA duplexes. In prokaryotes, the catabolite-activating protein, CAP, induces DNA bending of a similar magnitude when it binds to its specific binding site, as was described in Chapter 8.

Although all the nucleotides within the TATA box deviate in their conformations from regular B-DNA, there are two positions that are especially distorted. Between the first two and the last two base pairs of the TATA box there are very sharp kinks in the DNA. Between these kinks the DNA is smoothly curved and partially unwound. Two pairs of phenylalanine residues are partially inserted between the first two and the last two base pairs of the TATA box. These phenylalanine wedges prevent the stacking of the base pairs, keeping them apart, which causes a significant increase in the rise of the DNA at this base pair step, hence a severe kink. The kinks at each end of the TATA box, in combination with the partial unwinding of the DNA in between, produce a wide and shallow minor groove. This groove is exposed at the convex outer surface of the TATA box region and interacts intimately with the entire concave undersurface of the TBP saddle between the stirrups (see Figure 9.5). All three distortions of the TATA box—the bending of the DNA, the widening of the minor groove and the unwinding of the DNA—are mechanically coupled and overall there is no topological strain in the DNA.

Most sequence-specific regulatory proteins bind to their DNA targets by presenting an α helix or a pair of antiparallel β strands to the major groove of DNA. Recognition of the TATA box by TBP is therefore exceptional; it utilizes a concave pleated sheet protein surface that interacts with the minor groove of DNA. Since the minor groove has very few sequence-specific

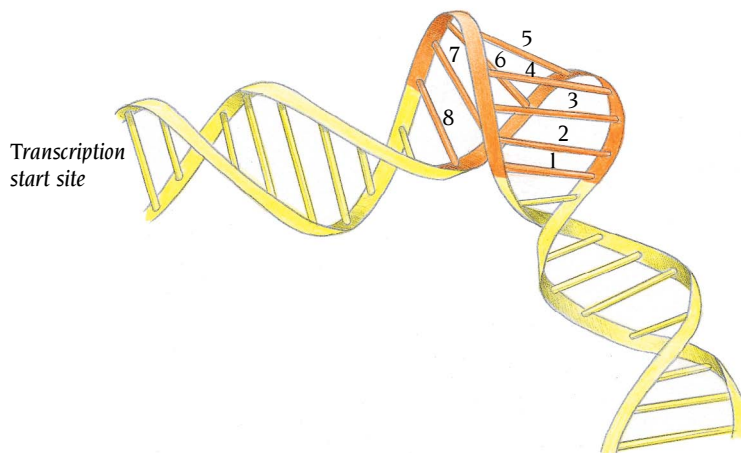


Figure 9.6 The DNA fragment is sharply bent in the TATA box region (orange) so that the DNA helices on both sides form an angle of about 110° to each other instead of 180° for undistorted B-DNA. The view of the DNA is rotated about 90° compared with that of Figure 9.5.

functional groups (see Chapter 7) it is appropriate to ask what factors account for the 100,000-fold preference of TBP for a TATA-box sequence over random DNA.

The interaction area between TBP and the TATA box is mainly hydrophobic

The interaction area between the underside of the TBP saddle and the minor groove of DNA is formed by two large, complementary surfaces with no water molecules between them. A surprisingly large amount of this area is hydrophobic, in contrast to protein–DNA interactions that involve the major groove, which are mainly hydrophilic and sometimes mediated by water molecules (see Chapter 8). In TBP, side chains from the eight central β strands interact with both the phosphate sugar backbone and the minor groove of the eight nucleotides of the TATA box. Fifteen side chains projecting from the β strands make hydrophobic contacts with the sugars and bases of DNA. The phosphate groups are hydrogen bonded to arginine and lysine side chains at the edges of the interaction area.

The only DNA sequence-specific aspect of these hydrophobic contacts is that they exclude G-C base pairs that are, of course, absent from the TATA box. The amino group of guanine (see Chapter 7) would, if present, sterically disrupt or modify the interface. Surprisingly, 12 of the 16 hydrogen bond acceptors of the minor groove (N3 of adenine and O2 of thymine) are buried in this interaction area without forming hydrogen bonds either to protein side chains or to water molecules. Studies of site-directed mutations have shown that burying polar side chains in a hydrophobic environment without satisfying their hydrogen-bonding requirements is energetically unfavorable. Clearly, the large number of energetically favorable hydrophobic interactions between TBP and TATA-box DNA must compensate for these unfavorable interactions in order to achieve binding constants in the nanomolar range.

The only sequence-specific hydrogen bonds between TBP side chains and the bases in the minor groove occur at the very center of the TATA box (Figure 9.7). The amide groups of two asparagine side chains donate four hydrogen bonds, two each to adjacent bases on the same DNA strand (Asn 69

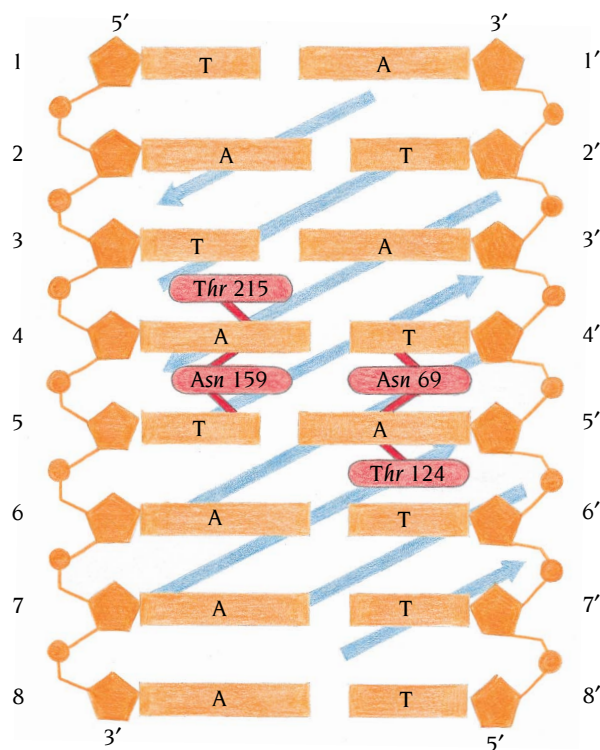


Figure 9.7 Sequence specific interactions between TBP and the TATA box. Asn 69 and Thr 124 from one domain and the equivalent residues Asn 159 and Thr 215 from the second domain interact with the palindromic TATA sequence of the central region of the TATA box.

to O2 of T4' and to N3 of A5', and Asn 159 to O2 of T5 and to N3 of A4). In addition, N3 of A5' and A4 accept hydrogen bonds from Thr 124 and Thr 215 respectively. There are also two conserved valine residues on each side of base pairs 4 and 5, Val 71 and 122 on one side and Val 161 and 213 on the other side. The side chains of the valine residues would cause steric interference with the NH₂ substituent from G-C or C-G base pair. The flanking valine residues in combination with the six hydrogen bonds specify A-T or T-A at positions 4 and 5 of the TATA box.

Like Thr 124 and Thr 215, the Asn 69 and Asn 159 residues occupy equivalent positions in the two homologous motifs of TBP. By analogy with the symmetric binding of a dimeric repressor molecule to a palindromic sequence described in Chapter 8, the two motifs of TBP form symmetric sequence-specific hydrogen bonds to the quasi-palindromic DNA sequence at the center of the TATA box. The consensus TATA-box sequence has an A-T base pair at position 4, but either a T-A or an A-T base pair at the symmetry-related position 5, and the sequence is, therefore, not strictly palindromic. However, the hydrogen bonds in the minor groove can be formed equally well to an A-T base pair or to a T-A base pair, because O2 of thymine and N3 of adenine occupy nearly stereochemically equivalent positions, and it is sufficient, therefore, for the consensus sequence of the TATA box to be quasi-palindromic.

In conclusion, one important factor that contributes to the strong affinity of TBP proteins to TATA boxes is the large hydrophobic interaction area between them. Major distortions of the B-DNA structure cause the DNA to present a wide and shallow minor groove surface that is sterically complementary to the underside of the saddle structure of the TBP protein. The complementarity of these surfaces, and in addition the six specific hydrogen bonds between four side chains from TBP and four hydrogen bond acceptors from bases in the minor groove, are the main factors responsible for causing TBP to bind to TATA boxes 100,000-fold more readily than to a random DNA sequence.

Functional implications of the distortion of DNA by TBP

The untwisting and strand separation of DNA are distortions necessary for transcription. These distortions occur at specific DNA sequences that contain a T-A step in the double helix, either in isolation, as described in Chapter 8 for CAP promoter regions in bacteria, or as the tandem repeat T-A-T-A in the TATA box of eucaryotic DNA. The B-DNA structure of such sequences is less stable than those containing G-C base pairs because the stacking interactions between A-T and T-A base pairs are relatively weak. TBP exploits this property of the TATA box to distort and bend the DNA and initiate transcription. The TATA box thus carries information for transcriptional initiation primarily at a physical level: its DNA is easily deformed. Chemical information, in the form of sequence-specific interactions between nucleotides and the protein, plays only a minor role, in contrast to the binding of specific transcription factors to the major groove of DNA. This is also reflected in the sequences of different TATA boxes, where the actual combinations of T-A and A-T base pairs are quite diverse. The minor groove recognition and the bendability of DNA works equally well for different combinations provided there are no G-C or C-G base pairs involved.

The sharp bend of DNA at the TATA box induced by TBP binding is favorable for the formation of the complete DNA control module; in particular, for the interaction of specific transcription factors with TFIID. Since these factors may bind to DNA several hundred base pairs away from the TATA box, and at the same time may interact with TBP through one or several TAFs, there must be several protein-DNA interactions within this module that distort the regular B-DNA structure (see Figure 9.2). The DNA bend caused by the binding of TBP to the TATA box is one important step to bring activators near to the site of action of RNA polymerase.

TFIIA and TFIIB bind to both TBP and DNA

TFIIA and **TFIIB** are two basal transcription factors that are involved in the nucleation stages of the preinitiation complex by binding to the TBP–TATA box complex. Crystal structures of the ternary complex **TFIIA**–TBP–TATA box have been determined by the groups of Paul Sigler, Yale University, and Timothy Richmond, ETH, Zurich, and that of the **TFIIB**–TBP–TATA box by Stephen Burley and collaborators. The TBP–DNA interactions and the distortions of the DNA structure are essentially the same in these ternary complexes as in the binary TBP–TATA complex.

TFIIB is arranged in two domains, both of which have the cyclin fold described in Chapter 6. Both domains bind to the TBP–TATA box complex at the C-terminal stirrup and helix of TBP. The phosphate and sugar moieties of DNA form extensive non-sequence-specific contacts with **TFIIB** both upstream and downstream of the middle of the TATA box.

TFIIA also has two domains, one of which is a four-helix bundle and the other an antiparallel β sandwich. The β sandwich interacts with the N-terminal half of TBP and thus positions **TFIIA** on the other side of the complex compared with **TFIIB**. This domain also interacts with phosphates and sugars of DNA upstream of the TATA box. The four-helix bundle domain makes no contact with DNA or TBP and is far removed from the position of **TFIIB**.

These structures have given us a first glimpse of the preinitiation complex for transcription by RNA polymerase II. Only TBP provides sequence-specific recognition of DNA. The transcription factors **TFIIA** and **TFIIB** bind to both TBP and DNA in such a way that they are separated from each other and provide large surfaces available for binding to other components of the preinitiation complex. The components of the preinitiation complex assemble on the promoters with high cooperativity. TBP exhibits no detectable affinity toward either **TFIIA** or **TFIIB** in the absence of DNA. Similarly, the promoter segment binds **TFIIA** and **TFIIB** poorly or not at all in the absence of TBP. However, the bend imposed on the promoter by TBP repositions the charges and functional groups on the DNA so that **TFIIA** and **TFIIB** bind with high affinity, anchored properly in the complex by specific interactions with TBP. In essence, the TBP–promoter complex is a nucleoprotein unit that binds the basal factors **TFIIA** and **TFIIB** with high affinity and specificity.

Homeodomain proteins are involved in the development of many eucaryotic organisms

Eucaryotes have many more genes and a broader range of specific transcription factors than procaryotes and gene expression is regulated by using sets of these factors in a combinatorial way. Eucaryotes have found several different solutions to the problem of producing a three-dimensional scaffold that allows a protein to interact specifically with DNA. In the next chapter we shall discuss some of the solutions that have no counterpart in procaryotes. However, the procaryotic helix–turn–helix solution to this problem (see Chapter 8) is also exploited in eucaryotes, in **homeodomain** proteins and some other families of transcription factors.

The **homeobox**, a DNA sequence of about 180 base pairs within the coding region of certain genes, was first discovered in the fruitfly *Drosophila* during studies of mutations that cause bizarre disturbances of the fly's body plan, so-called homeotic transformations. In the mutation *Antennapedia*, for example, legs grow from the head in place of antennae. Such homeotic mutations cause a whole set of cells to be misinformed as to their location in the organism and consequently to form a structure appropriate to another place. Homeoboxes have since been found in several hundred different genes from both vertebrates and invertebrates, and there are varying degrees of DNA sequence homology between different members of this superfamily. A number of these homeobox genes have retained both their precisely ordered tandem arrangement in the genome, and their developmental roles in axial patterning across more than 500 million years of evolution.

Homeoboxes code for homeodomains, sequences of 60 amino acids that function as the DNA-binding regions of transcription factors. Each homeobox gene in *Drosophila* is expressed only in its own characteristic subset of embryonic cells, and almost every embryonic cell contains a unique combination of homeodomain proteins.

Monomers of homeodomain proteins bind to DNA through a helix-turn-helix motif

Comparisons of the amino acid sequences of homeodomains with the sequences of procaryotic DNA-binding proteins suggested that part of the homeodomain forms a helix-turn-helix motif similar to the DNA-binding motif of Cro and repressor, described in Chapter 8. This suggestion was verified by the NMR determination of the solution structures of the *Antennapedia* homeodomain and its complex with a DNA fragment in the laboratory of Kurt Wüthrich, ETH, Zurich, and by an x-ray structure determination of a DNA complex with a different homeodomain from the *Drosophila* gene *engrailed*, in the laboratory of Carl Pabo at Johns Hopkins University, Baltimore. Subsequently, the x-ray structure of a DNA complex of the yeast Mat $\alpha 2$ repressor, which regulates the expression of cell-type specific genes in yeast, was determined by the group of Carl Pabo. The overall structures of these three complexes are almost identical, differing only in the details of their protein-DNA interactions.

The homeodomain is built up from three α helices, connected by rather short loop regions (Figure 9.8). The three helices are formed by residues 10 to 22, 28 to 38, and 42 to 59 (see Figure 9.11). Alpha helices 2 and 3 and the loop region between them form the helix-turn-helix motif, and the structure of the region between residues 30 to 50 is very similar to the helix-turn-helix motif of Cro and repressor. Indeed the three-dimensional structures of this region of these three homeodomains are as similar to the procaryotic helix-turn-helix motifs as they are to each other. The remaining regions of the homeodomain structure, however, differ from the structures of procaryotic DNA-binding proteins. Homeodomain helix 1 is differently arranged compared with the additional helices in λ or trp repressor, and homeodomain helix 3 is considerably longer than the recognition helices of the procaryotic helix-turn-helix motifs (Figure 9.9). Interestingly, the additional helical turns of helix 3 in the *Antennapedia* homeodomain are apparently induced by DNA binding because in the absence of DNA these residues are disordered.

The homeodomain frequently binds to DNA as a monomer, in contrast to procaryotic DNA-binding proteins containing the helix-turn-helix motif, which usually bind as dimers. *In vitro* the homeodomain binds specifically to

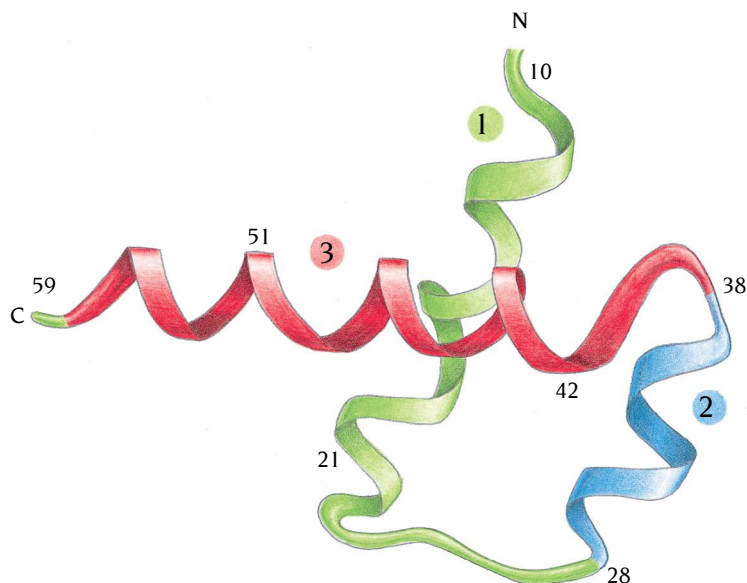


Figure 9.8 Schematic diagram of the three-dimensional structure of the *Antennapedia* homeodomain. The structure is built up from three α helices connected by short loops. Helices 2 and 3 form a helix-turn-helix motif (blue and red) similar to those in procaryotic DNA-binding proteins. (Adapted from Y.Q. Qian et al., *Cell* 59: 573–580, 1989.)

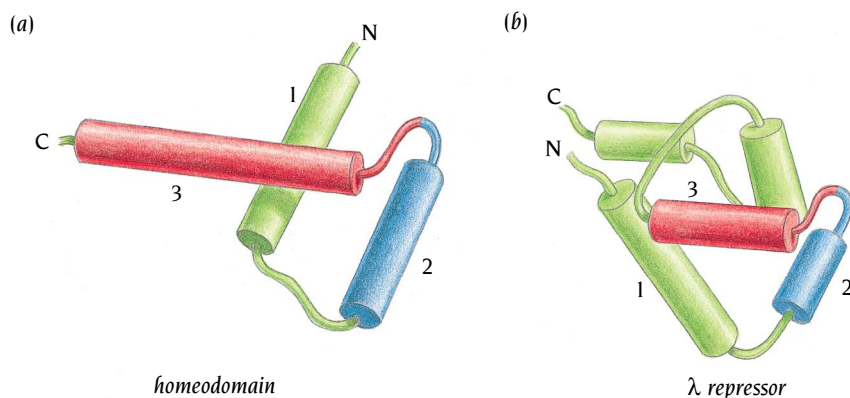


Figure 9.9 Comparison of the helix-turn-helix motifs in homeodomains (a) and λ repressor (b). The recognition helix (red) of the homeodomain is longer than in the procaryotic repressor motif. In addition the first helix of the homeodomain [(green in (a))] is oriented differently.

DNA fragments containing the sequence 5'-A-T-T-A-3' with a dissociation constant of about 1 nanomolar. It also binds nonspecifically to DNA fragments with different sequences, but with about 100 times lower affinity. In all three homeodomain structures the DNA fragment is essentially regular B-DNA. As shown in Figure 9.10, the recognition helix of the homeodomain is positioned in the major groove of the B-DNA and the extended polypeptide chain at the N-terminus, before helix 1, is positioned on the opposite side of the DNA, in the minor groove. The homeodomain-DNA complex is stabilized by a number of salt bridges and hydrogen bonds between the DNA backbone and protein side chains, involving mainly lysine and arginines, from four regions of the protein: the N-terminus, the loop between helices 1 and 2, helix 2 and the recognition helix 3 (Figure 9.11). The positions of the residues involved in these nonspecific DNA contacts vary slightly in these homeodomain complexes but the overall arrangement of the homeodomains bound to DNA is virtually identical. The amino acid sequence identity between the three is only 20%, and such remarkable conformational identity between such distantly related members of the homeodomain family strongly suggests that their conserved structure-function relationships have played important roles in the evolution of eucaryotic organisms.

Residues 3, 5, 6, and 8 in the N-terminal arm lie in the minor groove and form contacts with either the edge of the bases or with the DNA backbone. Almost all homeodomains contain four conserved residues, Asn 51, Arg 53, Trp 48 and Phe 49, in the middle of the long recognition helix. The first two conserved polar residues interact with DNA. The second two are part of the hydrophobic core of the homeodomain, and are important for the accurate positioning of the recognition helix and the N-terminal arm with respect to

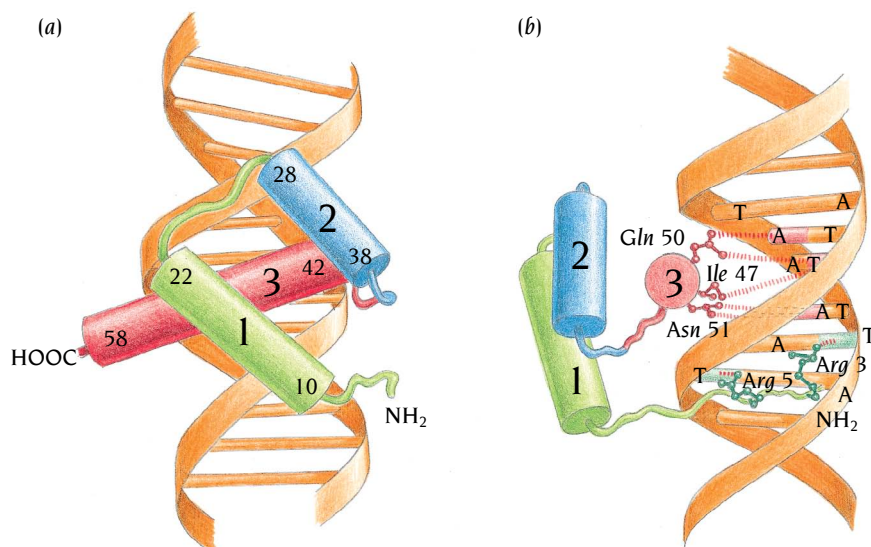


Figure 9.10 Schematic diagrams illustrating the complex between DNA (orange) and one monomer of the homeodomain. The recognition helix (red) binds in the major groove of DNA and provides the sequence-specific interactions with bases in the DNA. The N-terminus (green) binds in the minor groove on the opposite side of the DNA molecule and arginine side chains make nonspecific interactions with the phosphate groups of the DNA. (Adapted from C.R. Kissinger et al., *Cell* 63: 579-590, 1990.)

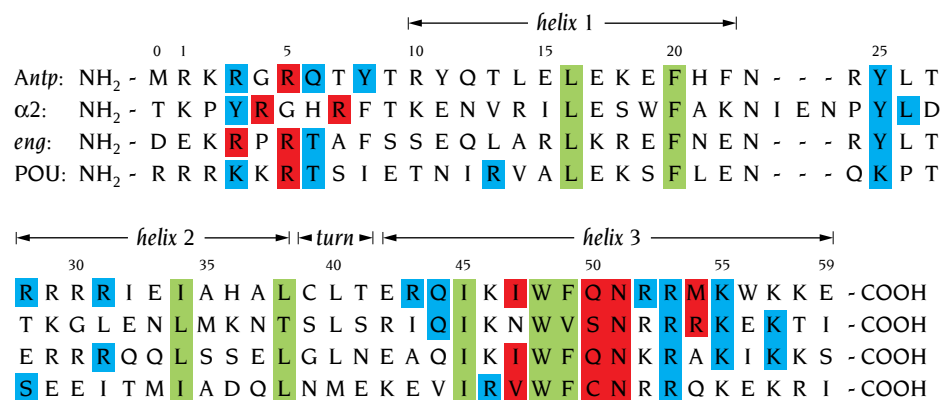


Figure 9.11 Amino acid sequences of homeodomains from four different transcription factors: Antp is from the *Antennapedia* gene in the fruitfly *Drosophila*, α2 is from the yeast *Mat α2* gene, eng is from the engrailed gene in *Drosophila* and POU is from the POU homeodomain in the mammalian gene *Oct-1*. Residues colored green form the hydrophobic core of the homeodomain, blue form nonspecific interactions with the DNA backbone and red form contacts with the edges of the DNA bases.

each other and to the remaining parts of the homeodomain structure. The side chains of Asn 51 and residues 47, 50, and 54 point into the major groove and make either direct or water-mediated contacts with the base edges of the DNA recognition site (see Figure 9.10). These contacts allow homeodomains to bind to operator sites containing the consensus sequence 5'-A-T-T-A-3', and also to some extent differentiate between different operator sites by interacting with flanking bases.

The NMR study by Wüthrich and coworkers has shown that there is a cavity between the protein and the DNA in the major groove of the *Antennapedia* complex. There are several water molecules in this cavity with a residence time with respect to exchange with bulk water in the millisecond to nanosecond range. These observations indicate that at least some of the specific protein-DNA interactions are short-lived and mediated by water molecules. In particular, the interactions between DNA and the highly conserved Gln 50 and the invariant Asn 51 are best rationalized as a fluctuating network of weak-bonding interactions involving interfacial hydration water molecules.

In vivo specificity of homeodomain transcription factors depends on interactions with other proteins

Homeodomains that have closely related amino acid sequences bind with comparable affinities to the same DNA sequence *in vitro*. But *in vivo* transcription factors with similar homeodomains must activate the expression of their different target genes in a highly selective and precise way to avoid, for example, catastrophic deformations of the developing body plan of an organism. How is this selective activity achieved? How do functionally different homeodomain proteins achieve the necessary selectivity and specificity when their DNA binding regions are virtually identical? The answer lies in the fact that homeodomain proteins, like other classes of transcription factors, do not operate alone but in concert with other transcription factors and associated proteins.

These additional proteins cooperate with the homeodomain to provide the required specificity for the target genes. For example combinations of the **Mat α2** homeodomain transcription factor with either of two different transcription factors **Mcm1** or **Mat a1** specify two different cell types in yeast. **Mat a1** has a homeodomain whereas **Mcm1** is a different type of transcription factor. **Mat α2** contains an N-terminal dimerization domain that binds to **Mcm1** and a C-terminal region that binds to **Mat a1** in each case to form a DNA-binding heterodimer. The crystal structure of the **Mat α2**-**Mat a1** heterodimer in a complex with DNA has been determined by the group of Cynthia Wolberger at Johns Hopkins University, and compared to the **Mat α2**-DNA complex.

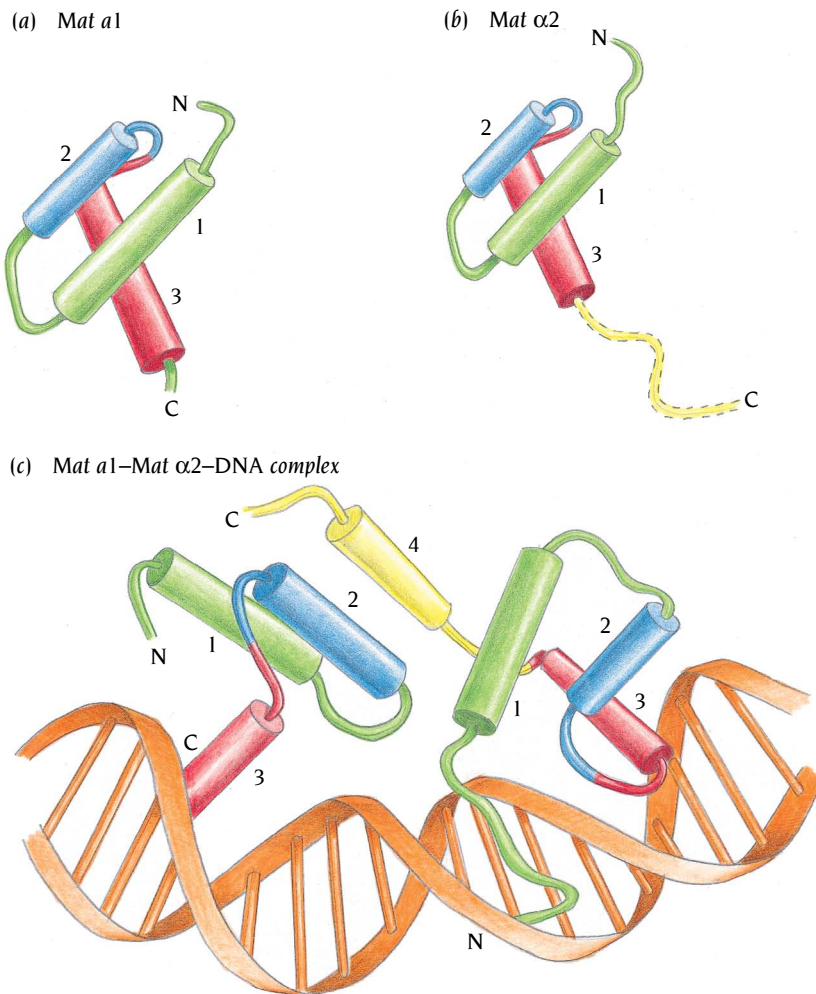


Figure 9.12 Schematic diagram of the structure of the heterodimeric yeast transcription factor Mat α 2–Mat α 1 bound to DNA. Both Mat α 2 and Mat α 1 are homeodomains containing the helix–turn–helix motif. The first helix in this motif is colored blue and the second, the recognition helix, is red. (a) The assumed structure of the Mat α 1 homeodomain in the absence of DNA, based on its sequence similarity to other homeodomains of known structure. (b) The structure of the Mat α 2 homeodomain. The C-terminal tail (dotted) is flexible in the monomer and has no defined structure. (c) The structure of the Mat α 1–Mat α 2–DNA complex. The C-terminal domain of Mat α 2 (yellow) folds into an α helix (4) in the complex and interacts with the first two helices of Mat α 1, to form a heterodimer that binds to DNA. (Adapted from B.J. Andrews and M.S. Donoviel, *Science* 270: 251–253, 1995.)

Mat α 1 and Mat α 2 bind in tandem to DNA with their recognition helices in the major groove (Figure 9.12). The C-terminal tail of Mat α 2 provides the only contacts with Mat α 1 in the heterodimer. A portion of the tail forms a short α helix with three leucine residues which project into a hydrophobic pocket between helices 1 and 2 of the Mat α 1 homeodomain. In the absence of Mat α 1 this tail is flexible with no ordered structure. The 21-base pair DNA in this complex has a significant overall bend of 60° , in contrast to the B-DNA in the Mat α 2–DNA complex. The DNA is smoothly bent without dramatic local distortion or kinks. Without the observed bend in the DNA the C-terminal tail of Mat α 2 would not be able to reach its binding site on Mat α 1. Proper dimerization and positioning of the two recognition helices consequently require the DNA to be bent and provide the necessary energy for the bending.

How is the binding specificity of the heterodimer achieved compared with the specificity of Mat α 2 alone? The crystal structure rules out the simple model that the contacts made between the Mat α 2 homeodomain and DNA are altered as a result of heterodimerization. The contacts between the Mat α 2 homeodomain and DNA in the heterodimer complex are virtually indistinguishable from those seen in the structure of the Mat α 2 monomer bound to DNA. However, there are at least two significant factors that may account for the increased specificity of the heterodimer. First, the Mat α 1 homeodomain makes significant contacts with the DNA, and the heterodimeric complex will therefore bind more tightly to sites that provide the contacts required by both partners. Second, site-directed mutagenesis experiments have shown that the protein–protein interactions involving the

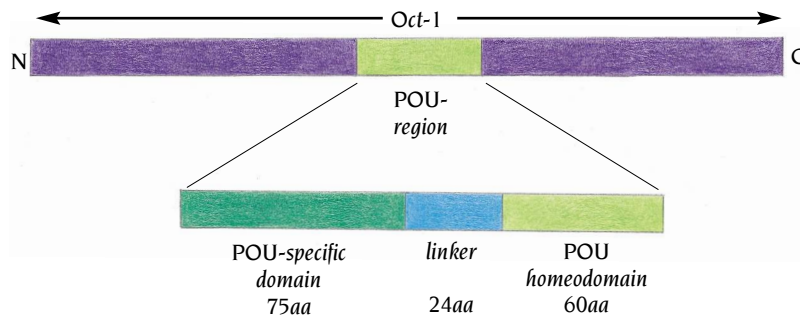


Figure 9.13 The DNA-binding region of the protein Oct-1, the POU region (green), comprises two domains, the POU-specific domain (dark green) and the POU homeodomain (light green) joined by a linker region (blue). These two domains bind to DNA in a tandem arrangement.

C-terminal tail of Mat $\alpha 2$ and the Mat $\alpha 1$ homeodomain are crucial for the specificity of DNA binding. The heterodimer interface may increase specificity by dictating the precise spacing of the two binding sites for the recognition helices, analogous to the 434 Cro and repressor proteins discussed in Chapter 8. This could be one way in which regions of transcription factors outside the DNA-binding domains, the so-called activating regions, provide specificity of transcriptional activation *in vivo*.

POU regions bind to DNA by two tandemly oriented helix-turn-helix motifs

A tandem arrangement of two DNA-binding domains, called the POU region, is present in a class of transcription factors involved in development and in the expression of growth factors, histones and immunoglobulins. The POU region is a 150–160 amino acid sequence consisting of a homeodomain and a POU-specific domain joined by a short variable linker region (Figure 9.13). NMR studies by the groups of Robert Kaptein in Utrecht University and Peter Wright at The Scripps Research Institute, La Jolla, established that the POU-specific domain has a structure very similar to the λ repressor, in spite of having no apparent sequence homology; as described in Chapter 8, this structure is very simple, consisting of four α helices where helices 2 and 3 form a helix-turn-helix motif. The POU region has, therefore, two helix-turn-helix motifs that are similar while the remaining structures of its two domains are quite different.

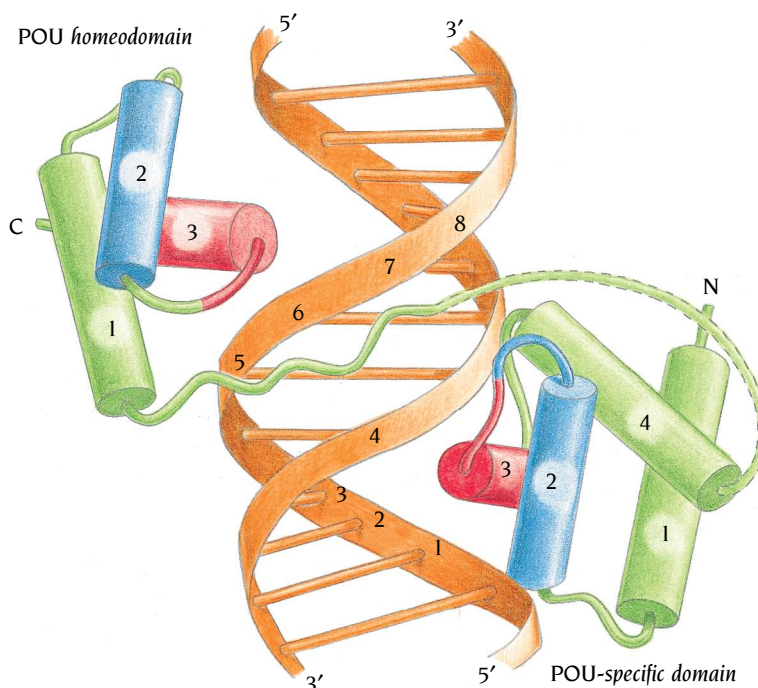


Figure 9.14 The two domains of the POU region bind in tandem on opposite sides of the DNA double helix. Both the POU-specific domain and the POU homeodomain have a helix-turn-helix motif (blue and red) which binds to DNA with their recognition helices (red) in the major groove. The linker region that joins these domains is partly disordered. (Adapted from J.D. Klemm et al., *Cell* 77: 21–32, 1994.)

The x-ray structure of a DNA complex with the complete POU region from the transcription factor Oct-1, which regulates transcription of small nuclear RNA genes and the histone H2B gene, was subsequently determined by the group of Carl Pabo. The Oct-1 POU region recognizes the octamer nucleotide sequence 5'-A-T-G-C-A-A-T-3' *in vivo*, hence the name Oct-1. Pabo found that the two helix-turn-helix domains bind in tandem along the octamer DNA sequence (Figure 9.14). The λ repressor-like POU-specific domain makes its essential contacts with the 5' half of the octamer site, and the POU homeodomain makes its contacts with the AAAT sequence in the 3' half. These two domains bind on opposite sides of the DNA double helix with their recognition helices in the major groove. There are no protein-protein contacts between the POU-specific domain and the POU homeodomain, and the 24-residue long linker region between the domains is disordered. This tandem arrangement of the two domains increases the binding specificity of the POU region to the octamer site above the specificities of the individual domains. Such a tandem arrangement of DNA-binding domains is also utilized by the "classic" zinc fingers to enhance their specificity, as will be described in Chapter 10.

Both domains of the POU region bind to DNA by the usual combination of non specific binding to the DNA backbone and specific binding to the bases. The contacts between the homeodomain and DNA are similar to those of the engrailed homeodomain (compare Figures 9.10b and 9.15a) and the

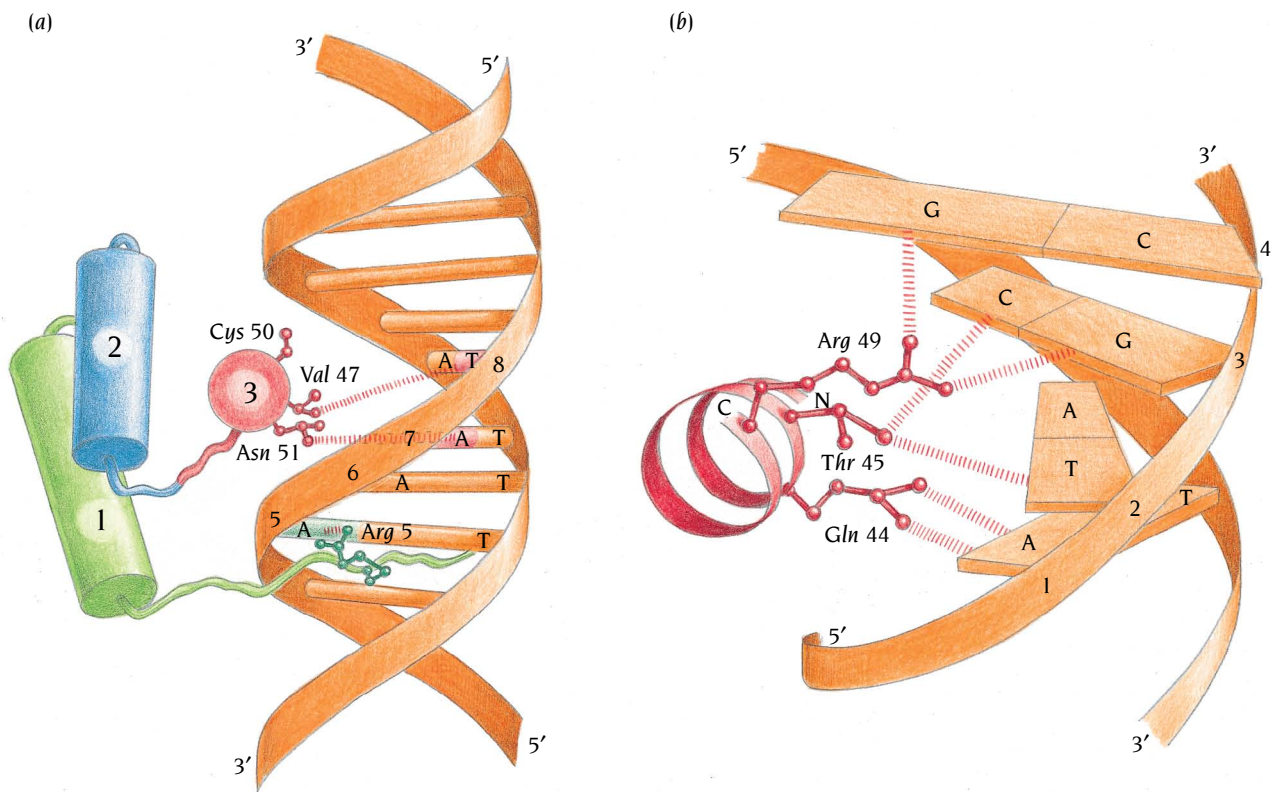


Figure 9.15 Diagram illustrating the sequence-specific contacts between DNA and the POU region. (a) Contacts from the POU homeodomain. Three residues from the recognition helix, Val 47, Cys 50 and Asn 51, interact with base pairs 7 to 8 in the major groove of the second half of the POU-binding site. In addition, Arg 5 in the N-terminal arm interacts with base pair 5 of the DNA in the minor groove. (b) Contacts from the POU-specific region. Three residues from the recognition helix, Gln 44, Thr 45 and Arg 49, interact with bases in the major groove of the first half of the POU-binding site, base pairs 1 to 4. These contacts are from the opposite side of the DNA molecule compared with those in (a). The numbering system starts from the first residue of the homeodomain. (Adapted from J.D. Klemm et al., *Cell* 77: 21–32, 1994.)

contacts between DNA and the POU-specific domain are almost identical to those observed in the λ repressor–DNA complex (Figure 9.15b). Helix 3 of the POU-specific domain is positioned in the major groove and makes all the base contacts with the ATGC subsite. The first residue of this helix, Gln 44, makes two hydrogen bonds to the adenine at base pair 1 and is also involved in a network of hydrogen bonds involving Arg 20, Gln 27, Glu 51 and the DNA backbone. These four residues (Arg 20, Gln 27, Gln 44 and Glu 51) are conserved in all of the over 20 different POU-specific domain sequences known, and identical residues are found at corresponding positions of the 434 repressor– and 434 Cro–DNA complexes. Such similarities raise the possibility that the eucaryotic POU-specific domains are evolutionarily related to bacteriophage repressors.

Much remains to be learnt about the function of homeodomains in vivo

Although homeobox genes are easy to identify from their nucleotide sequences, the target genes for homeodomain proteins have been difficult to locate because we lack knowledge about the additional factors required for the sequence specificity of these transcription factors *in vivo*. Which gene products could be involved? When tissues are formed in multicellular organisms, different types of cells are brought into contact with each other through complex cellular interactions, and signals are exchanged between cells leading to organized programs of cell division and cell differentiation. Prime candidate proteins involved in these complex interactions are cell adhesion molecules such as tenacin and cadherin, as well as growth factors and associated proteins involved in signal transduction, some of which we shall discuss in Chapter 13. How homeodomain proteins regulate the expression of such molecules is a key element of one of the central questions in molecular cell biology: how do cells form tissues during embryonic development, producing the characteristic body plan of a species?

Understanding tumorigenic mutations

A protein with the innocuous name **p53** is one of the most frequently cited biological molecules in the Science Citation Index. The “p” in p53 stands for protein and “53” indicates a molecular mass of 53 kDa. The p53 protein plays a fundamental role in human cell growth and mutations in this protein are frequently associated with the formation of tumors. It is estimated that of the 6.5 million people diagnosed with one or another form of cancer each year about half have p53 mutations in their tumor cells and that the vast majority of these mutations are single point mutations.

Wild-type p53 inhibits tumor formation, and its gene is therefore called a tumor suppressor gene. Details of exactly how this suppression is accomplished are currently unknown, but one important role of p53 is to maintain the integrity of the genome during cell division by controlling a critical step in the cell cycle. A complex series of molecular interactions regulates the cell cycle; the chief players are proteins called cyclins and their associated enzymes, cyclin-dependent kinases, which we discussed in Chapter 6. These kinases are inhibited by another protein, p21, whose expression is promoted by p53. In the presence of p21 the cell cycle is halted before the cell is committed to divide. Presumably this gives the cell time either to repair damaged DNA or, if it is beyond repair, to initiate programmed cell death (apoptosis).

One of the most important molecular functions of p53 is therefore to act as an activator of p21 transcription. The wild-type protein binds to specific DNA sequences, whereas tumor-derived p53 mutants are defective in sequence-specific DNA binding and consequently cannot activate the transcription of p53-controlled genes. As we will see more than half of the over one thousand different mutations found in p53 involve amino acids which are directly or indirectly associated with DNA binding.

The monomeric p53 polypeptide chain is divided in three domains

The polypeptide chain of p53 is divided in three domains, each with its own function (Figure 9.16). Like many other transcription factors, p53 has an N-terminal activation domain followed by a DNA-binding domain, while the C-terminal 100 residues form an oligomerization domain involved in the formation of the p53 tetramers. Mutants lacking the C-terminal domain do not form tetramers, but the monomeric mutant molecules retain their sequence-specific DNA-binding properties *in vitro*.

Intact p53 has so far resisted crystallization, and it is too large for current NMR technology, so the three-dimensional structure of the complete molecule is not known. However, the group of Nikola Pavletich at the Memorial Sloan-Kettering Cancer Center in New York has determined the crystal structure of the DNA-binding domain (residues 102–292) bound to a 21-base pair DNA fragment containing a specific p53-binding sequence. Furthermore, the same group has determined the x-ray structure of a region of the oligomerization domain, residues 325–356, involved in the formation of tetramers. These structural studies have enhanced our understanding of tumorigenic mutants of p53, and they give a solid basis both for mutational studies of p53 and for the design of cancer drugs.

The oligomerization domain forms tetramers

The x-ray studies have shown that the 32-amino acid peptide comprising residues 325–356 of the oligomerization domain of p53 is sufficient for tetramer formation. The structure of each unit is very simple; a β -strand-turn- α -helix motif (Figure 9.17). Two such units form a dimer through interactions between the β strands and the α helices. The β strands form an antiparallel two-stranded β sheet and the two α helices are arranged in an antiparallel fashion. The interactions that hold the subunits together as dimers are mainly provided through eight backbone hydrogen bonds in the two-stranded β sheet. In addition there is a small interior core of hydrophobic side chains from both the β strands and the α helices.

Two such dimers form the tetramer through mainly hydrophobic interactions between the α helices. The β strands are on the outside of the tetramer and are not involved in the dimer–dimer interactions. The arrangement of the four α helices is unusual and provides a rare example of four α helices packed against each other in a way different from the four-helix bundle motif.

Although most of the p53 mutations detected in tumors are located in the DNA-binding domain, a few have also been observed in the oligomerization domain. The structure provides a clear explanation of the effects of two such mutations. One is a mutation of the β -strand residue Leu 330 to His. The side chain of this Leu residue in the wild-type protein is in the center of the hydrophobic core of the dimer; replacing it with the hydrophilic histidine side chain of the mutant destabilizes the core, prevents formation of dimers, and therefore inhibits p53 function. The second example is the mutation of a glycine residue in the turn between the β strand and the α helix. The turn comprises only this glycine residue, which adopts an unusual conformation in a region of the Ramachandran plot (see Figure 1.7) that is energetically

Figure 9.17 Schematic diagram illustrating the tetrameric structure of the p53 oligomerization domain. The four subunits have different colors. Each subunit has a simple structure comprising a β strand and an α helix joined by a one-residue turn. The tetramer is built up from a pair of dimers (yellow-blue and red-green). Within each dimer the β strands form a two-stranded antiparallel β sheet which provides most of the subunit interactions. The two dimers are held together by interactions between the four α helices, which are packed in a different way from a four-helix bundle. (Adapted from P.D. Jeffrey et al., *Science* 267: 1498–1502, 1995.)

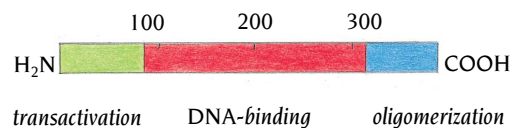
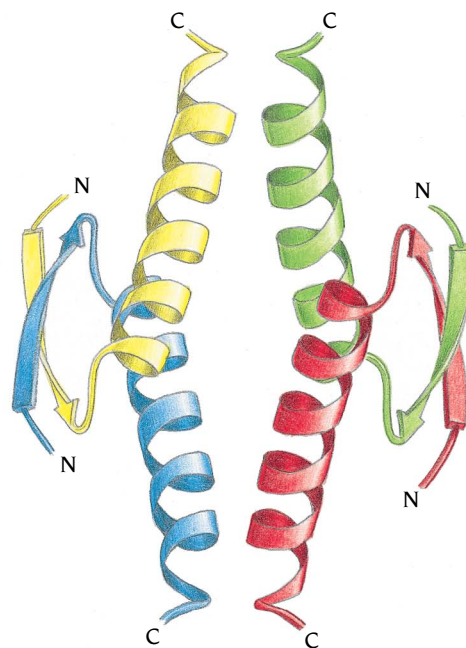
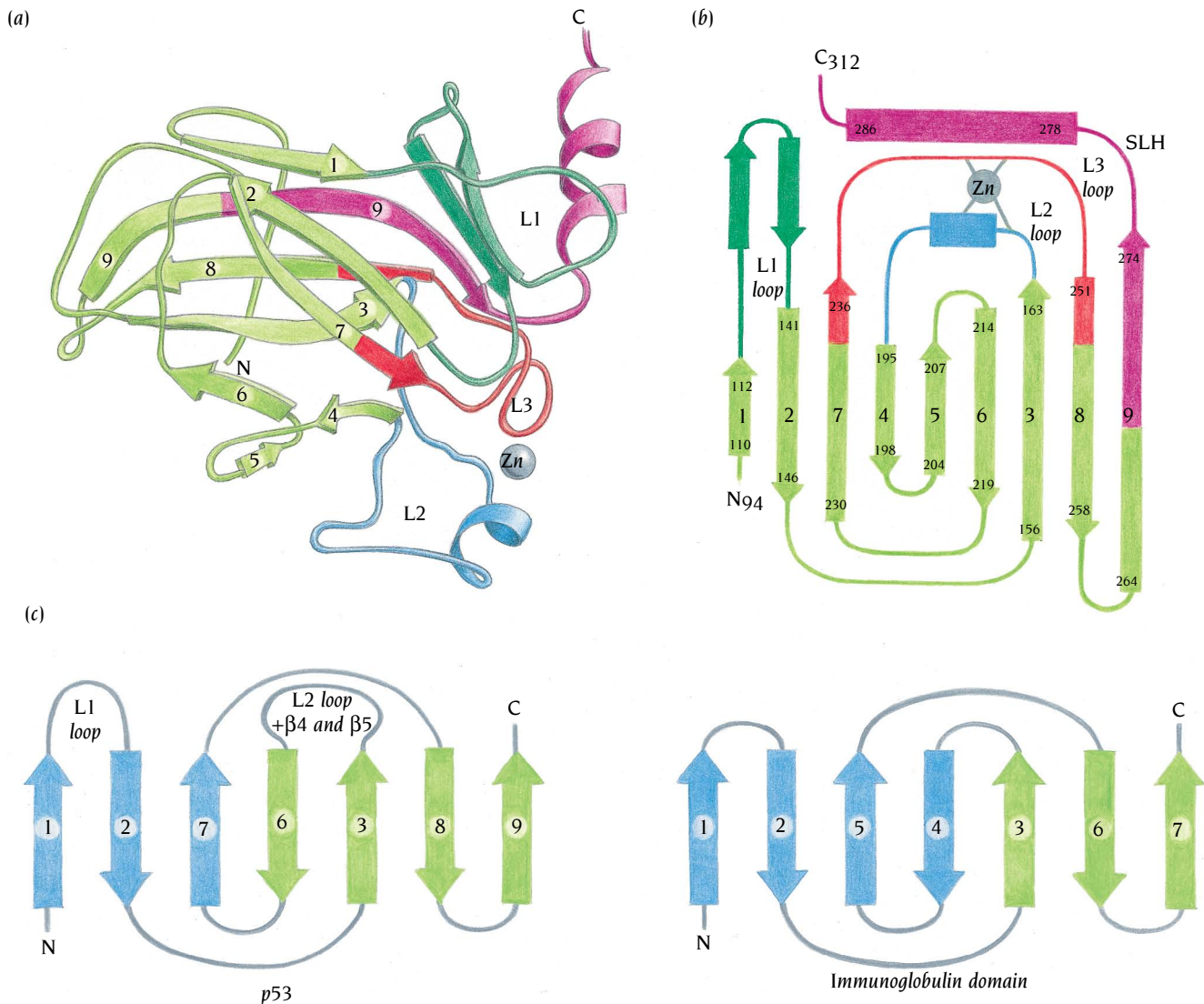


Figure 9.16 The polypeptide chain of the protein p53 is divided into three domains with different functions: transactivation, DNA binding and oligomerization.





unfavorable for any other residue than glycine. Mutation of this glycine residue prevents the subunit from folding into the correct structure for dimerization and hence abolishes p53 function.

The DNA-binding domain of p53 is an antiparallel β barrel

Transcription factors frequently utilize a few standard motifs for DNA binding, for example zinc-fingers or helix-turn-helix motifs, but the DNA-binding domain of p53 has none of these. Instead it binds to DNA using protruding loops that are anchored to an antiparallel β -barrel scaffold. The fold of this barrel is similar to the immunoglobulin fold which is the basic domain structure of immunoglobulins, as will be discussed in Chapter 15. The β barrel of p53 comprises nine β strands, seven of which form the immunoglobulin fold (Figure 9.18). From the ribbon diagram in Figure 9.18a it can be seen that β strands 4 and 5 are short and at the edges of the β sheets that build up the compressed barrel. The remaining seven strands of the barrel have not only the topology of an immunoglobulin fold (Figure 9.18c) but also an overall structure that is similar to members of the immunoglobulin superfamily, such as one domain of the MHC-binding coreceptor CD4, which also acts as a receptor for HIV.

p53 is not the only transcription factor having the immunoglobulin fold; it also occurs in the mammalian transcription factor NF- κ B, which regulates the expression of a number of genes involved in the response to infection

Figure 9.18 Schematic diagram of the structure of the DNA-binding domain of p53. (a) The DNA binding domain of p53 folds into an antiparallel β barrel with long loop regions—L1 (dark green), L2 (blue) and L3 (red)—at one end of the molecule. The conformations of loops L2 and L3 are stabilized by a zinc atom. The C-terminus of the chain (purple) is at the same end of the molecule as these loops. (b) Topological diagram of the DNA-binding domain of p53. Nine antiparallel β strands form a β barrel, the central part of which has a fold similar to that of immunoglobulin domains. A region called SLH (strand-loop-helix) spans residues 271–286. (c) Simplified topological diagram of p53 compared with a topological diagram of the constant immunoglobulin domain (see Figure 15.7). Strands of the same color belong to the same β sheet. Strand numbers 4 and 5 are considered to be part of the loop between strands 3 and 6. The middle strand in these diagrams belongs to different sheets but in the actual structures these strands are at the edges of the two sheets, in only slightly different positions relative to the common structure. [(b) Adapted from Y. Cho et al., *Science* 265: 346–355, 1994.]

1	5	10	15	20																
A	T	A	A	T	T	G	G	G	C	A	A	G	T	C	T	A	G	G	A	A
≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡
A	T	T	A	A	C	C	C	G	T	T	C	A	G	A	T	C	C	T	T	T

Figure 9.19 Nucleotide sequence of the 21-base pair DNA fragment cocrystallized with the DNA-binding domain of p53. The p53 binds in a sequence-specific manner to the shaded region.

and stress, including immunoglobulin genes. NF- κ B is a member of a family of transcription factors that is characterized by a homologous sequence of about 280 amino acid residues, called the REL-homology region. The groups of Steve Harrison and Paul Sigler determined the structure of the REL-homology region from NF- κ B and showed that it has two immunoglobulin domains that both use loop regions to bind DNA. Even though p53 does not have a REL-homology region in its amino acid sequence, it is a member of the family of transcription factors that use loop regions of the immunoglobulin fold for DNA binding.

At one end of the barrel the β strands are close together, forming a compact structure with short loop regions connecting the strands. The β strands are highly twisted and their directions diverge as they reach the other end of the barrel. Consequently the barrel is more open at this end, the loops are longer and they extend outside the barrel; this is the end of the molecule to which DNA binds (see Figure 9.20). The conformations of two of these loops are stabilized by a zinc atom, which is bound to two cysteine side chains from one loop and to one cysteine and one histidine side chain from the other. There is no structural resemblance to the zinc finger domains discussed in the next chapter, in spite of the similar zinc ligands: these ligands only reflect the preference of zinc to form strong bonds to nitrogen- and sulfur-containing side chains.

Two loop regions and one α helix of p53 bind to DNA

Figure 9.19 shows the sequence of the DNA that was used for the structure determination of the p53–DNA complex; the bases involved in sequence-specific binding to the protein are shaded. One molecule of the DNA-binding domain of p53 binds to the minor and the major grooves of the DNA making sequence-specific interactions with both strands (Figure 9.20).

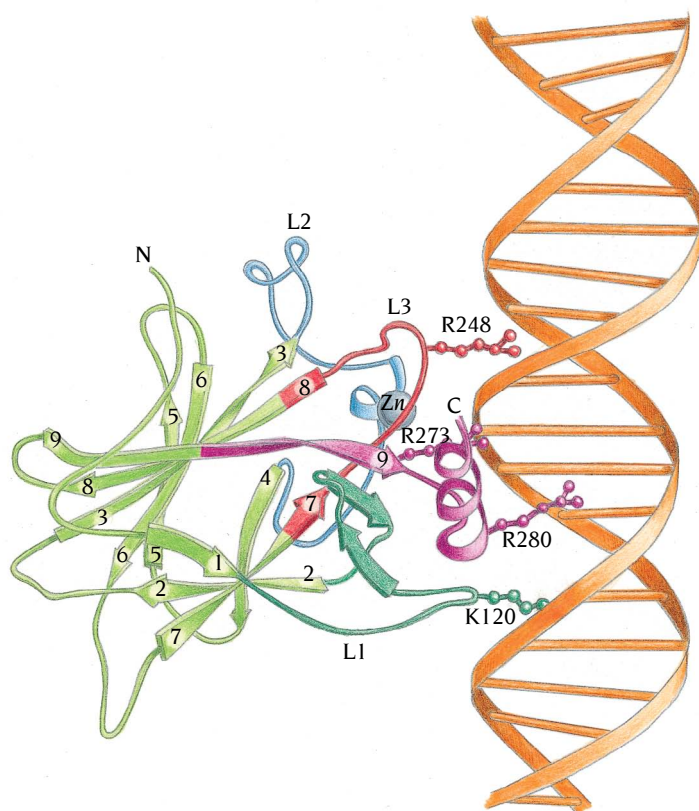


Figure 9.20 Diagram illustrating the sequence-specific interactions between DNA and p53. The C-terminal α helix and loop L1 of p53 bind in the major groove of the DNA. Arg 280 from the α helix and Lys 120 from L1 form important specific interactions with bases of the DNA. In addition, Arg 248 from loop L3 binds to the DNA in the minor groove. (Adapted from Y. Cho et al., *Science* 265: 346–355, 1994.)

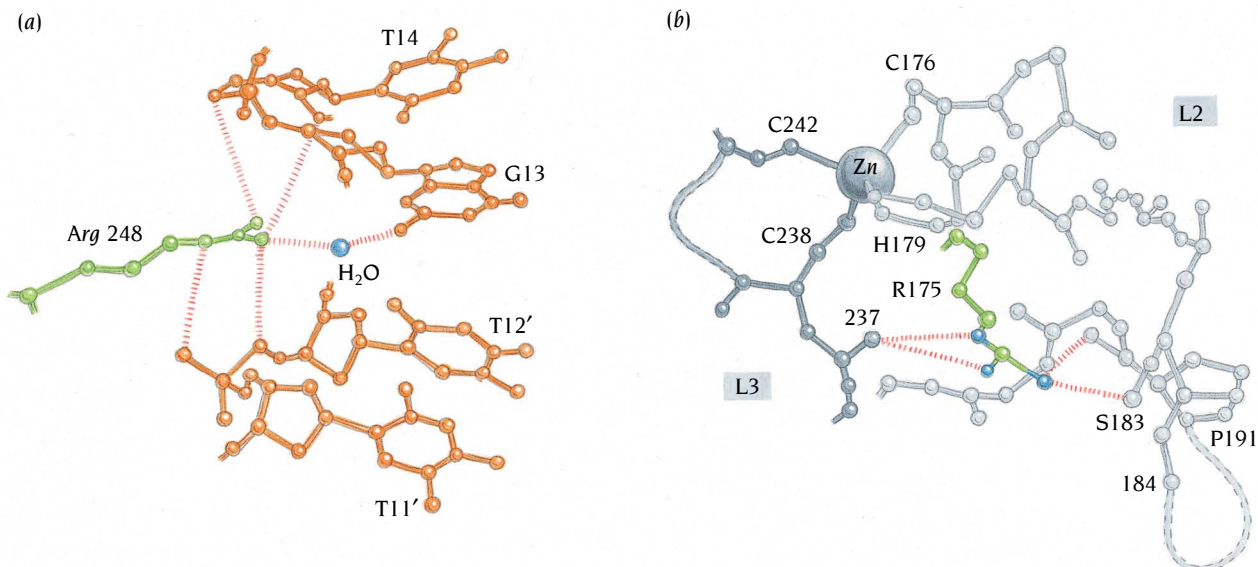


Figure 9.21 Detailed views of two arginine residues of p53 that are frequently mutated in tumors. (a) Interactions between Arg 248 (green) and the DNA (orange). The side chain of Arg 248 is wedged into the minor groove and makes contacts with the sugar and phosphate groups of T12' and T14. In addition, a water molecule (blue) mediates a hydrogen bond between Arg 248 and the base of G13. (b) Interactions between L2 and L3 in p53. The zinc atom is bound to two cysteine residues in L3 and to one cysteine and one histidine in L2. In addition, residue Arg 175 from L2 forms two hydrogen bonds to the main chain C=O group of residue 237 in L3 and one hydrogen bond to the side chain of Ser 183 in L2. Mutations in residues 175, 183 or 237 would distort loop L3 and prevent proper binding of p53 to DNA.

Interactions that are not sequence specific are also an important part of the binding and occur between the sugar and phosphate residues of the DNA and the side-chain and main-chain atoms of the protein. In the crystals the DNA fragment retains the B-DNA structure with only minor distortions.

Two of the four loop regions (L1 and L3) at the open end of the barrel are directly involved in DNA binding. In addition, the COOH terminus of the polypeptide chain extends from the barrel and forms a protruding α helix that is positioned in the major groove of the DNA. Residues from this α helix as well as from the preceding loop (from β strand 9) participate in both specific and nonspecific binding. The most critical of these sequence-specific contacts in the major groove of DNA is made by Arg 280 from the α helix, which donates hydrogen bonds to the base of G10' (see Figure 9.20). Because the C-G 10 base pair (see Figure 9.19) is invariant in all p53-specific binding sites and several point mutations of Arg 280 have been observed in tumors, we can infer that this contact is important for the proper function of p53. Loop L1, which connects the first two strands of the barrel, interacts with a second area of the major groove. The side chain of Lys 120 forms hydrogen bonds with the base of G8 (see Figure 9.20), and its main-chain NH group interacts with one phosphate oxygen atom.

The minor groove interactions occur in the AT-rich region (base pairs 11 and 12) and involve Arg 248 from loop L3, which connects β strands 7 and 8 (Figures 9.20 and 9.21a). The side chain of Arg 248 is wedged into the minor groove and makes contact with the sugar and phosphate groups of T12' and T14, as well as with a water molecule that in turn is hydrogen bonded to the base G13. These interactions introduce slight distortions of the B-DNA structure, by compressing the minor groove. This narrowing of the groove allows Arg 248 to pack tightly against the sugar phosphate groups inside the minor groove. It is likely that an AT-rich sequence is required for this compression to occur.

Tumorigenic mutations occur mainly in three regions involved in DNA binding

By examining some of the over one thousand tumor-causing point mutations of p53 in the light of its structure, we can identify features of p53 that are necessary for tumor suppression. The amino acids most frequently changed in cancer cells are at or near the protein–DNA interface; residues that are infrequently mutated, if at all, are in general far from the DNA-binding site.

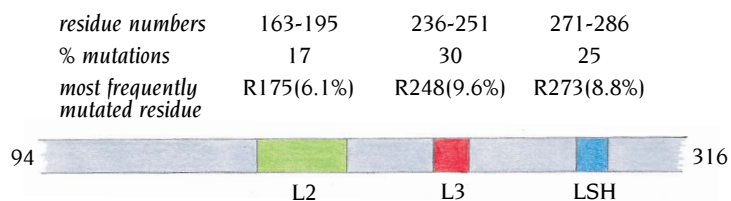


Figure 9.22 Most tumorigenic mutations of p53 are found in the regions of the polypeptide chain that are involved in protein–DNA interactions. These regions are loops L2 (green) and L3 (red) and a region called LSH (blue) which comprises part of β strand 9 as well as the C-terminal α helix.

Mutations are particularly common in three regions of the structure (Figure 9.22). One of these regions is at the C-terminus (residues 271–286), and includes part of β strand 9, the C-terminal α helix and the loop between them. This region is called SLH for strand-loop-helix (see Figure 9.18). The other two regions are loop L3 (residues 236–251), which interacts with the minor groove of DNA, and loop L2 (residues 163–195), which is wedged between L3 and the β barrel and stabilizes the position of L3.

Thirty percent of the tumor-derived mutations are in L3, which contains the single most frequently mutated residue, Arg 248. Clearly the interaction between DNA and the specific side chain of an arginine residue inside the minor groove is of crucial importance for the proper function of p53. It is an open question whether this interaction is needed for the recognition of specific DNA sequences, or is required for the proper distortion of the DNA structure, or a combination of both. Other residues that are frequently mutated in this region participate in interactions with loop L2 and stabilize the structures of loops L2 and L3. Mutations of these residues presumably destabilize the structure so that efficient DNA binding can no longer take place.

Although the L2 loop does not interact directly with DNA, the numerous interactions between both main-chain atoms and the side chains of loops L2 and L3 are necessary to keep L3 in the correct position for DNA binding. The importance of the interactions between loops L2 and L3 is illustrated by Arg 175 in L2 (Figure 9.21b). The side chain of Arg 175 forms two hydrogen bonds to the main-chain C=O group of residue 237 in L3 and one hydrogen bond to the side-chain oxygen atom of Ser 183 in L2. It is therefore an essential residue for maintaining the structural stability of both L2 and L3, and in fact it is the most tumorigenic residue in L2, being mutated in 6.1% of p53-induced tumors. Another example of the importance of loop L2 for function of p53 is provided by mutations that affect the binding of the zinc atom. The zinc atom that stabilizes the structure of loops L2 and L3 is bound to two side chains from L2 (Cys 176 and His 179) and to two side chains from L3 (Cys 238 and Cys 242) (Figure 9.21b). The crystal structure shows that mutations in these four important residues would distort loop L3 and prevent efficient binding to DNA, and such mutations are indeed frequently found in tumors.

Some of the residues from the SLH region bind directly to DNA in the major groove as described earlier (see Figure 9.20). The most frequently mutated residue in this region, Arg 273, binds to the phosphate group of T11'. In addition it participates in an extended network of hydrogen bonds with other side chains of the SLH region, including Arg 280, which makes the critical sequence-specific DNA interaction with the base of G10'. Mutations of Arg 280 are found in 2.1% of p53-induced tumors.

In conclusion it is obvious from a comparison of the structure of the DNA-binding domain of p53 and the pattern of p53 mutations observed in tumors that mutations that disturb or abolish specific p53 DNA-binding also abolish the ability of p53 to suppress tumors. The mutated residues are of two different types; those that participate directly in DNA binding and those in the two loop regions that are important for the overall structural integrity of the DNA-binding regions. The function of p53 may be particularly sensitive to the second class of mutations. The challenge for designers of anticancer drugs is to try to synthesize molecules that can stabilize these loop structures in mutant p53 molecules, thereby restoring the protein's normal DNA-binding function.

Conclusions

The general transcription factor TBP sharply bends and unwinds TATA box DNA, causing the minor groove of DNA to open up and become wide and shallow enough to accommodate a β sheet. Hydrophobic interactions occur between the side chains of the β sheet residues and sugars and bases of the DNA. The mechanism of DNA binding by TBP is therefore very different from that of the helix-turn-helix homeobox transcription factors that, like Cro and repressor in bacteriophages, form hydrophilic interactions with the major groove of DNA, in a base-specific manner. However, unlike Cro and repressor, which bind as homodimers, homeobox factors bind as monomers. In addition to binding in the major groove, the homeodomain proteins also exhibit minor groove base recognition. TBP and the helix-turn-helix transcription factors, like all other transcription factors, form salt bridges between their lysine and arginine side chains and phosphate groups of the DNA to stabilize the DNA:protein complex. The monomeric POU transcription factors have two tandemly oriented helix-turn-helix motifs in the same polypeptide chain, which bind to two sequential binding sites in the major groove of the DNA.

DNA binding by TBP is strongly dependent on the presence of T-A base pairs in the TATA box. Bending allows remote sites on the DNA, with their bound cognate specific transcription factors, to come close together such that the proteins can interact to form the transcriptional preinitiation complex.

The DNA-binding domains of the tetrameric tumor suppressor protein, p53, has an antiparallel β barrel scaffold with an immunoglobulin fold, from one end of which projects two loop regions that are directly involved in binding DNA in the major groove. In addition, the carboxy terminus of the DNA-binding domain extends from the barrel as an α helix and is also involved in binding to the major groove of DNA. Most of the tumorigenic point mutations of p53 are in these DNA-binding regions. Transcription factors such as NF- κ B that contain REL-homology regions also use the immunoglobulin fold as scaffold for loops that bind DNA.

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Specific Transcription Factors Belong to a Few Families

10

Many procaryotic transcription factors are dimers that recognize palindromic DNA sequences in such a way that each subunit binds one half-site. Dimerization and DNA-binding activities usually reside in different domains or regions of the protein, as was discussed in Chapter 8. The POU transcription factors exhibit a variation of this theme by having two similar domains in one polypeptide chain (see Chapter 9). A similar arrangement is found in another important class of specific eucaryotic transcription factors—those that have zinc atoms in their DNA-binding motifs. As we will see, the polypeptide chains of these proteins may contain several homologous domains, each capable of specific DNA binding and each containing zinc as an integral part of the DNA-binding domain. These proteins belong to several different groups with different structures and different modes of DNA binding. We will discuss three of them that illustrate three different ways by which zinc-containing motifs recognize specific binding sites arranged contiguously along DNA segments.

As we have already mentioned in Chapters 8 and 9, one unifying concept to emerge from the study of different DNA-binding motifs is that these motifs provide three-dimensional scaffolds that match the contours of DNA. These scaffolds ensure proper positioning of the interacting protein surfaces against the DNA, allowing both sequence-specific and nonspecific interactions to occur. In some cases it is the amino acid sequences of the scaffolds, rather than the sequences of amino acids involved in DNA interactions, that make it possible to identify a DNA-binding motif in a protein sequence. This principle applies in particular to two important families of eucaryotic transcription factors that contain leucine zippers. Later in this chapter we will discuss one example each from these two families: GCN4, which belongs to the basic region/leucine zipper (b/zip) family, and Max, which is a member of the basic region/helix-loop-helix/leucine zipper (b/HLH/zip) family. The HLH motif, which is also present in many transcription factors that do not have the leucine zipper region, is very different from the helix-turn-helix motif that is so frequent in procaryotic DNA-binding proteins (see Chapter 8) and in homeodomains (see Chapter 9).

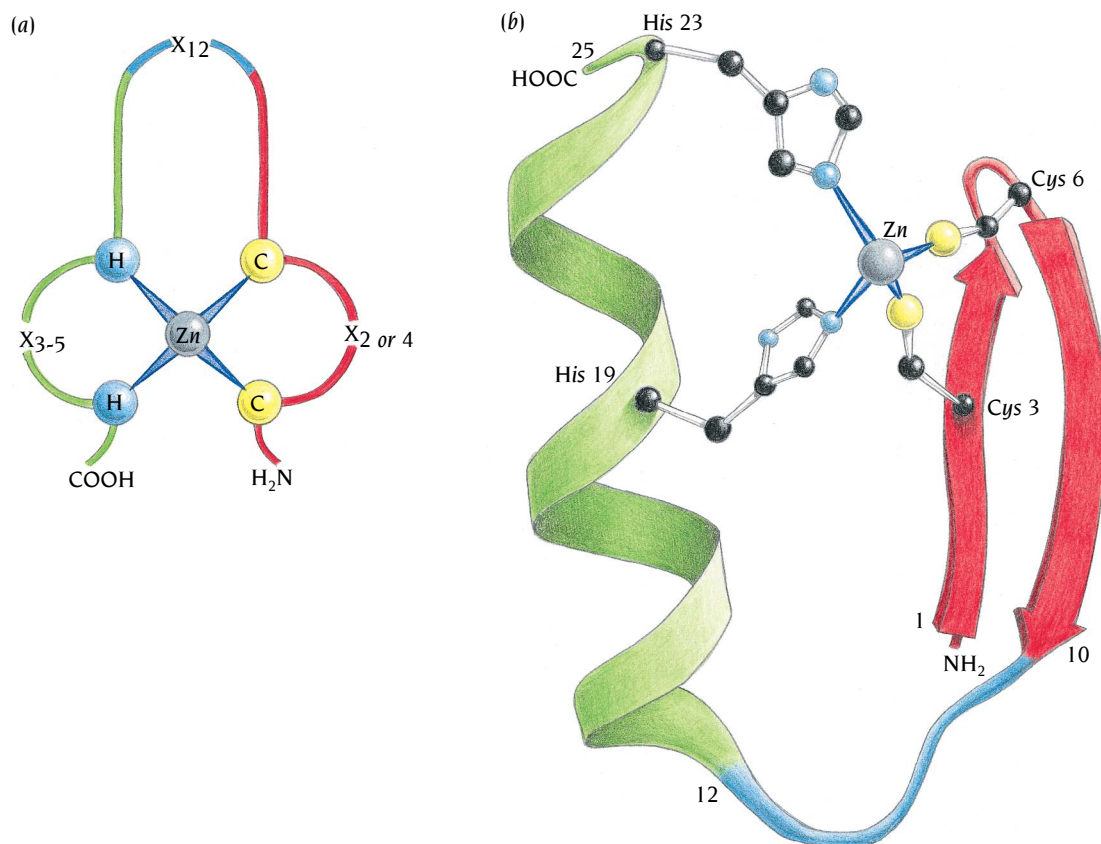
The DNA-binding motifs discussed in this and the preceding two chapters are those most frequently found in procaryotes and eucaryotes. However, other motifs are known, for example the β sheet motif of the met repressor in *Escherichia coli* which binds to the major groove of DNA. No doubt others remain to be discovered.

Several different groups of zinc-containing motifs have been observed

More than a thousand different transcription factors contain zinc as an essential element of their DNA-binding domains. The polypeptide chains of such zinc-containing motifs are usually short, about 50 amino acids or less, with regular patterns of cysteine and/or histidine residues along the chains. These residues bind to the zinc atoms, thereby providing a scaffold for the folding of the motif into a small compact domain. Several different subgroups of such motifs have so far been discovered, each with a different pattern of zinc-binding residues and with different three-dimensional structures.

The classic zinc finger motif was first described in 1985 in the laboratory of Aaron Klug at the MRC Laboratory of Molecular Biology in Cambridge, UK, where it was inferred from an analysis of the amino acid sequence of the transcription factor TFIIIA from *Xenopus laevis*. This factor, which regulates transcription of ribosomal 5S RNA, can be isolated in large quantities from frog oocytes. Each oocyte contains about 20,000 molecules of TFIIIA complexed with 5S RNA to form 7S ribonucleoprotein particles. The amino acid sequence of the 344-residue polypeptide chain of TFIIIA contains nine repeated sequences of about 30 residues each. The repeats are not identical in sequence but each contains two Cys residues at the amino end and two His residues at the carboxy end. Since the protein contains intrinsic zinc atoms and its transcriptional activity is dependent on the presence of these zinc atoms, Klug suggested that the Cys and His atoms are ligands to a zinc atom and that the loop between these residues forms the DNA-binding region. Each of the nine repeats was therefore called a zinc finger. As we shall see, these deductions were subsequently confirmed by structural studies. Since 1985, thousands of different genes coding for similar motifs have been identified and sequenced from a variety of eucaryotic organisms. They all contain two cysteine residues separated by two or four amino acids and two histidine residues separated by three to five amino acids (Figure 10.1a). The linker region between the last cysteine and the first histidine is 12 residues long.

Figure 10.1 (a) The classic zinc finger motif comprises about 30 amino acid residues, with two cysteine and two histidine residues which bind to a zinc atom. The linker region between the last cysteine and the first histidine is 12 residues long and is called the finger region. (b) Schematic diagram of the three-dimensional structure of a chemically synthesized 25-residue peptide with an amino acid sequence corresponding to one of the zinc fingers in an embryonic protein, Xfin, from *Xenopus laevis*. The structure is built up from an antiparallel β hairpin motif (residues 1 to 10) followed by an α helix (residues 12 to 24). The four zinc ligands Cys 3, Cys 6, His 19, and His 23 anchor one end of the helix to one end of the β sheet. Models, quite similar to the observed structure, were predicted from amino acid sequences of members of this zinc finger family. (Adapted from M.S. Lee et al., *Science* 245: 635–637, 1989.)



The classic zinc finger is a motif that is repeated in tandem to recognize DNA sequences of different lengths, with each finger interacting with a small number of base pairs. Other zinc finger motifs bind as monomers to discrete sites or as dimers to palindromic DNA sequences, and frequently several binding sites in the DNA are separated by hundreds of base pairs in the transcriptional control region. The strength of the interaction can be varied by changes in the sequence of both the protein and the DNA and by varying the length of the spacing between the fingers. These changes allow a high level of specificity in recognition, and the modular design offers a large number of combinatorial possibilities for the specific recognition of DNA.

Many attempts have been made to crystallize the transcription factor TFIIIA, but so far no crystals have been obtained that diffract to high resolution. However, the structures of a number of other individual classic zinc fingers have been determined. The first were obtained in the laboratories of Rachel Klevit, University of Washington, Seattle, and Peter Wright at the Scripps Clinic, La Jolla, from chemically synthesized peptides. Wright used a peptide corresponding to one zinc finger of a protein from *Xenopus laevis*, encoded by an embryonic gene called *Xfin* (*X* for *Xenopus*, *fin* for fingers). This protein has no less than 37 tandemly repeated classic zinc fingers.

It has also been possible to determine the x-ray structures of classic zinc finger motifs from several proteins bound to specific DNA fragments. We will here describe one such structure containing three zinc fingers from a mouse protein, Zif 268, which is expressed at an early developmental stage of the mouse. Nikola Pavletich and Carl Pabo at the Johns Hopkins University School of Medicine, Baltimore, determined the x-ray structure to 2.1 Å resolution of a recombinant polypeptide derived from Zif 268 bound to a 10-base

Figure 10.2 (a) Amino acid sequence of a fragment of the Zif 268 protein that contains three zinc fingers. Residues forming the β strands and α helices are red and green, respectively, and those involved in the turn between the last β strand and the α helix are blue. (b) The nucleotide sequence of the DNA fragment that was used in the x-ray structure determination of the Zif 268 fragment complexed with DNA.



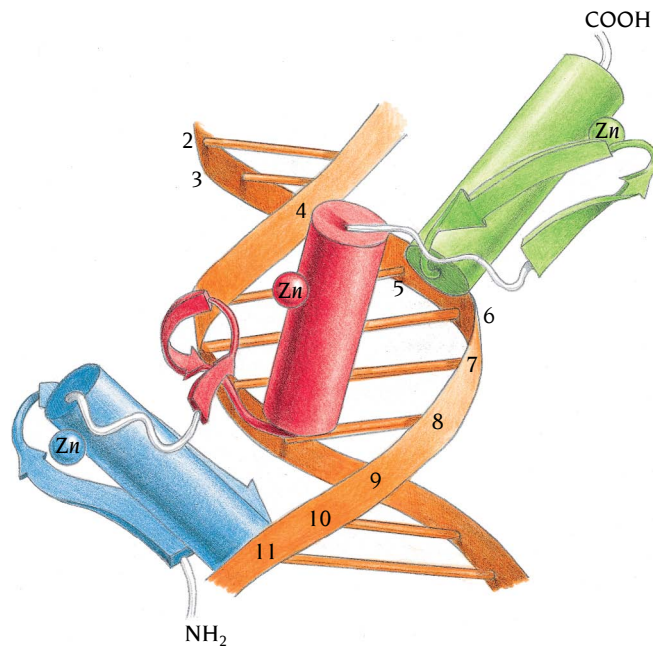


Figure 10.3 Schematic diagram of the structure of three zinc fingers of Zif 268 bound to DNA. The three zinc fingers, which bind in tandem to the major groove of DNA, are colored blue, red and green from the N-terminus. The zinc fingers have the same structure and bind in a similar way with the N-terminus of the α helix pointing into the major groove. (Adapted from N.P. Pavletich et al., *Science* 261: 1701–1707, 1993.)

pair DNA fragment, with a sequence corresponding to the consensus sequence of Zif 268 binding sites (Figure 10.2b). The 90-residue Zif 268 polypeptide comprises three zinc finger motifs joined by linker regions each of four residues (Figure 10.2a). The structures of the three zinc finger motifs in the Zif 268–DNA complex are very similar to each other and are essentially the same as that of the Xfin motif shown in Figure 10.1b, while the DNA fragment in the complex is essentially undistorted B-DNA. Complex formation therefore does not appreciably change the structure of either the zinc finger motifs or the DNA.

The DNA fragment binds to the polypeptide with a dissociation constant of 6 nanomoles, indicating tight binding, and the overall structure of the complex reveals why tandemly repeated zinc fingers are such efficient motifs for protein–DNA recognition. The three zinc fingers are arranged in a semi-helical structure that follows the major groove of the B-DNA (Figure 10.3). The helix of each finger fits directly into the major groove, with residues from the N-terminal region of each helix in contact with base pairs in the major groove. Each of the three zinc fingers uses its helix in a similar fashion and each finger makes its sequence-specific contacts with a subsite of three base pairs. The three tandemly oriented zinc fingers therefore bind to nine consecutive base pairs in the major groove. Although the α helix fits into the major groove, it is tilted with respect to the groove such that the C-terminal region of the helix is pointing away from the bases and does not make contacts with DNA. The β sheet is on the other side of the helix, away from the base pairs, and is shifted toward one side of the major groove. The two strands of the β sheet have very different roles in the complex. The first does not make any contacts with the DNA whereas the second β strand contacts the sugar-phosphate backbone along one strand of the DNA.

The finger region of the classic zinc finger motif interacts with DNA

The 12 residues between the second cysteine zinc ligand and the first histidine ligand of the classic zinc finger motif form the “finger region”. Structurally, this region comprises the second β strand, the N-terminal half of the helix and the two residues that form the turn between the β strand and the helix. This is the region of the polypeptide chain that forms the main interaction area with DNA and these interactions are both sequence specific,

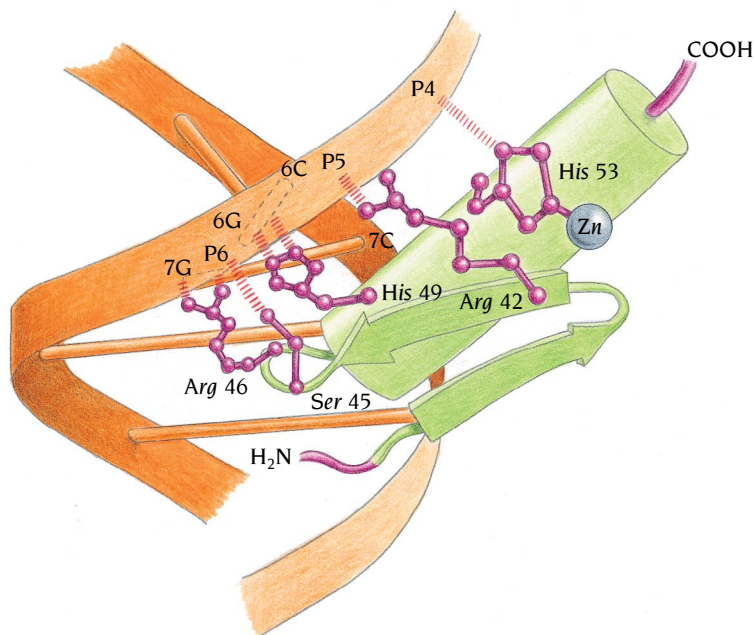


Figure 10.4 Detailed view of the binding of the second zinc finger of Zif 268 to DNA. Two side chains, Arg 46 and His 49, form sequence-specific interactions with DNA. There are also three nonspecific interactions between phosphate groups of the DNA and the side chains of Arg 42, Ser 45, and His 53.

between side chains of the protein and the bases of DNA, and nonspecific, mainly between the phosphate oxygen atoms of DNA and side chains of the protein.

A typical example of these interactions, for the second zinc finger of the Zif 268–DNA complex, is shown in Figure 10.4. Five side chains from the polypeptide bind to DNA, two of them in a sequence-specific manner. Arg 46, which is the last residue in the loop region and immediately precedes the helix, forms two hydrogen bonds with the guanine base of the G–C base pair 7. His 49, which is the third residue of the helix, forms hydrogen bonds with the guanine base of the G–C base pair 6. In addition to these sequence-specific hydrogen bonds, there are three nonspecific interactions between phosphate oxygen atoms of base pairs 4, 5, and 6 and the side chains of His 53, Arg 42, and Ser 45, respectively. The participation of His 53 in interactions with DNA is noteworthy, since this side chain is also one of the zinc ligands. The role of this side chain is therefore twofold: to stabilize the finger structure by providing a zinc ligand from one of the nitrogen atoms of the imidazole ring, and to participate in DNA binding by forming a hydrogen bond from the second nitrogen atom of the ring.

The same general picture of zinc finger binding to DNA observed in the Zif 268–DNA complex has also been found in two other DNA complexes formed by classic zinc fingers from the proteins GLI and TTK. GLI is the product of a human oncogene and TTK is a transcriptional control regulator of the *Drosophila* developmental gene *fushi-tarazu*. The GLI–DNA structure contains five zinc finger motifs, of which fingers 4 and 5 are specifically bound to DNA. The TTK–DNA structure contains two zinc fingers, both of which are specifically bound to DNA. The arrangement of these fingers in the major groove of DNA is very similar to the arrangement of two consecutive zinc fingers in the Zif 268–DNA complex. In all cases, the specific DNA contacts are made from the amino terminal half of the helix. This helix is oriented around its axis in a similar way in all zinc fingers, such that the side chain immediately preceding the first His ligand, residue 6 of the helix, is facing the edge of the bases in DNA. Consequently helix residues 2 and 3 are also facing the bases, as is the last residue of the loop region. Due to the geometry of the α helix, these four residues are all on the same side of the helix. Almost all sequence-specific contacts observed in these structures involve different combinations of these four residues (Figure 10.5). The few exceptions involve the remaining three residues of this N-terminal region of the helix.

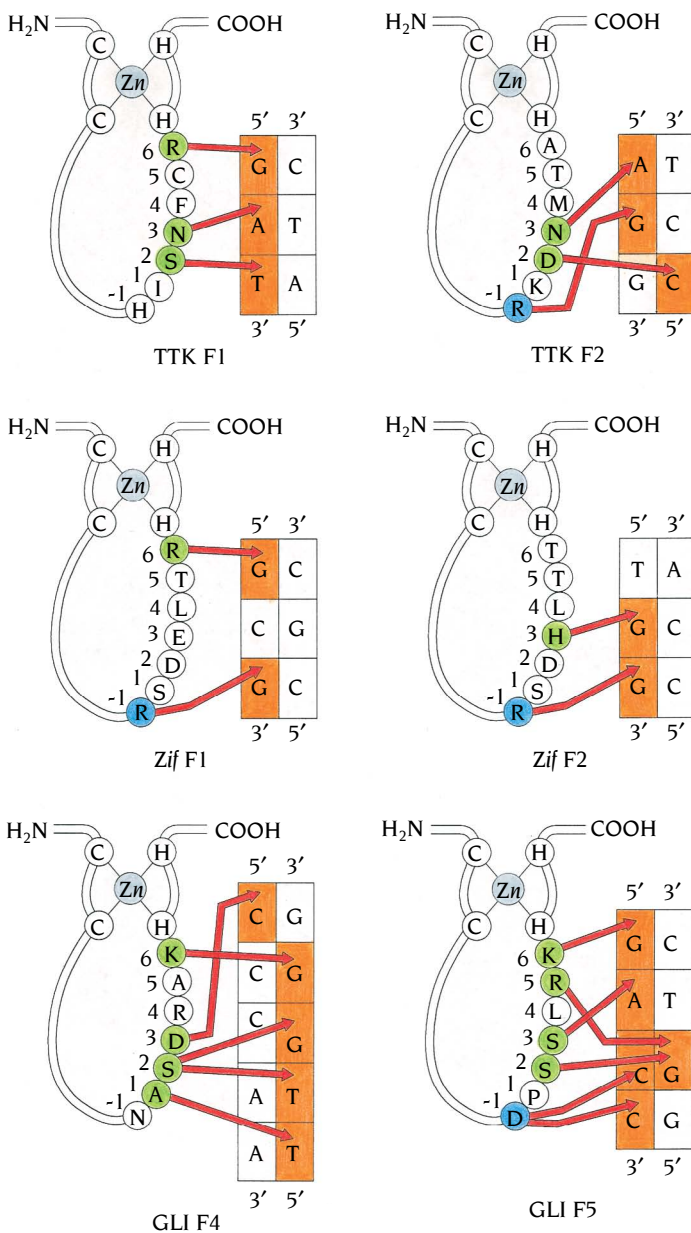
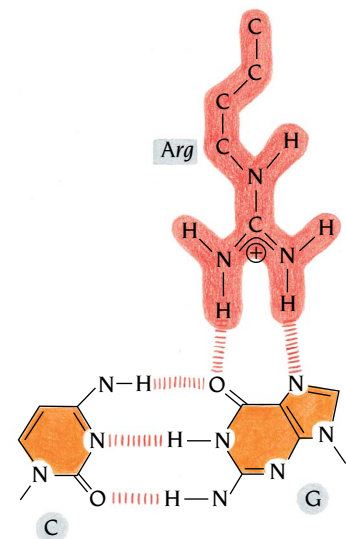


Figure 10.5 Comparison of the sequence-specific binding to DNA of six different zinc fingers. Residues in the N-terminus of the α helix in the finger regions are numbered 1 to 6. The residue immediately preceding the α helix is numbered -1. Amino acid residues and nucleotides that make sequence-specific contacts are colored. In spite of the structural similarities between the zinc fingers and their overall mode of binding, there is no simple rule that governs which bases the fingers contact.

The comparison of sequence-specific contacts shown in Figure 10.5 reveals that there is no specific correspondence between an amino acid residue in one of the four critical positions of the α helix and the base that it contacts. Although contiguous zinc fingers contact contiguous sets of base pairs, there is no simple rule that governs which bases the fingers contact. Furthermore, the number of bases in each set varies from three to five, and sometimes the sets of bases are strictly contiguous while in other cases they overlap or are spaced by one base pair that is not in contact with any finger. In addition, contacts are made in a seemingly random fashion to bases of both DNA strands. It is therefore not possible at present to predict from the amino acid sequence of a zinc finger the DNA sequence to which it will bind, or vice versa. However, one sequence-specific interaction occurs more

Figure 10.6 One sequence-specific interaction occurs more frequently than others in protein–DNA complexes: two hydrogen bonds form between an arginine side chain of the protein and a guanine base of the DNA, as shown in this diagram.



frequently than the others: that between an arginine side chain and a guanine base (Figure 10.6). The two strong hydrogen bonds that can be formed between these two groups make an essential contribution to sequence-specific recognition, as in the other families of DNA-binding proteins discussed in Chapters 8 and 9. The nonspecific interactions anchor the zinc fingers into the major groove of the DNA and are important for the tandem arrangement of the fingers along the groove. The interactions occur mainly from the linker regions between the zinc finger motifs and from side chains of the second β strand, which is aligned along one of the backbone chains of the DNA. Furthermore, the hydrogen bond from the histidine zinc ligand, exemplified above in the zinc finger 2 of the Zif 268–DNA complex, is found in most zinc finger–DNA interactions.

In conclusion, the structures of the classic zinc finger–DNA complexes reveal a simple and efficient mechanism for recognizing specific sites on double-stranded DNA. The α helix of each zinc finger fits directly into the major groove of B-DNA and side chains from the N-terminal region of this helix contact the edges of the base pairs. Most sequence-specific interactions involve side chains from the same four positions of the helix and arginine–guanine contacts occur frequently. The zinc fingers are tandemly arranged in the major groove and successive zinc fingers bind to successive subsites of three to five base pairs along the DNA.

Two zinc-containing motifs in the glucocorticoid receptor form one DNA-binding domain

The glucocorticoid receptor is a member of a family of transcription factors that includes the thyroid hormone receptor, the retinoic acid receptors, the vitamin D₃ receptor and different steroid hormone receptors (Figure 10.7). All members of this family have a highly conserved DNA-binding domain containing zinc that consists of about 70 residues and that binds to activating elements of DNA called hormone-response elements. In addition, they all contain a variable C-terminal ligand-binding domain, and some of them also have a large N-terminal domain, involved in transcriptional activation.

Protein fragments, produced from recombinant DNA and containing the glucocorticoid receptor DNA-binding domain, exhibit sequence-specific DNA binding to glucocorticoid response elements. These protein fragments contain two zinc atoms that are required for proper folding and DNA binding. The three-dimensional solution structures of several such receptor fragments have been determined by NMR methods. The first, from the glucocorticoid receptor and from the estrogen receptor, were determined in the laboratories of Robert Kaptein, University of Utrecht, and of Aaron Klug, University of Cambridge, UK, respectively.

The amino acid sequence of the glucocorticoid fragment indicated that there are two zinc-containing regions, each binding a zinc atom through four cysteine residues instead of two Cys and two His residues (Figure 10.8a). In the three-dimensional structure these two zinc-binding motifs (red and green

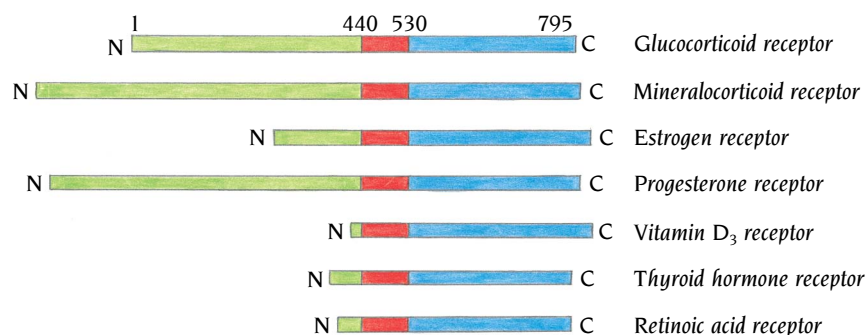


Figure 10.7 Evolutionarily related members of the receptor family of transcription factors that have four cysteine residues bound to zinc in each of two regions of the sequence. The DNA-binding domains (red) have highly homologous amino acid sequences, whereas the ligand-binding domains (blue) are more variable. Residue numbers of the domain boundaries are given for the glucocorticoid receptor. Exchange of individual domains between different members of the family suggests that they can function as interchangeable modules.

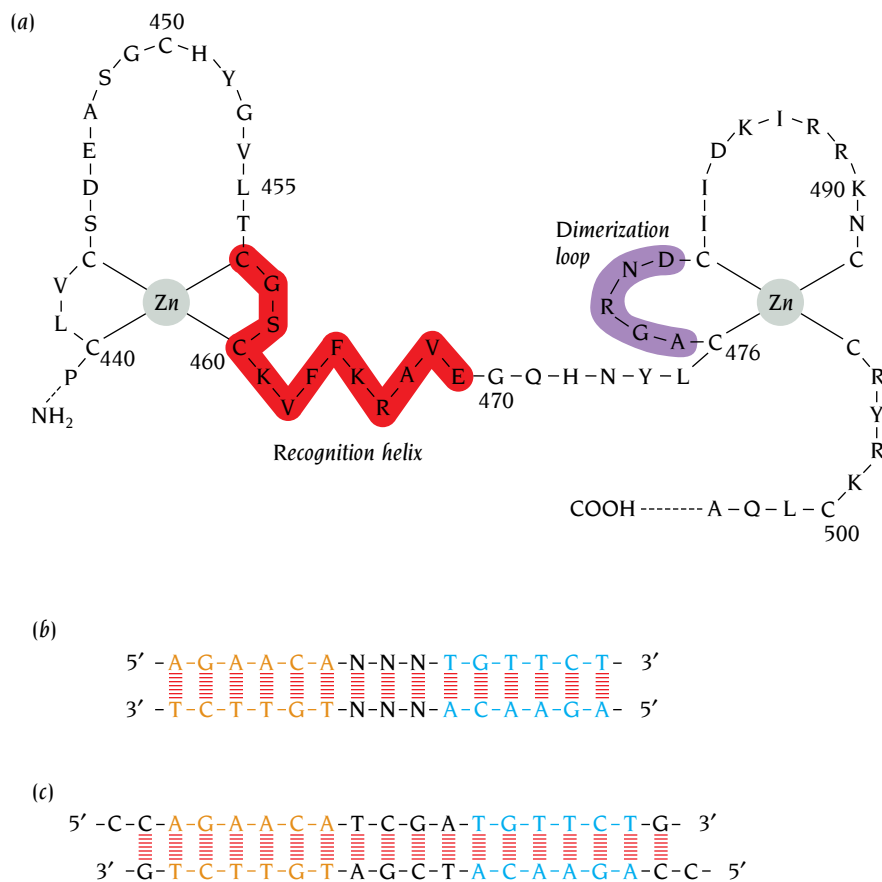


Figure 10.8 (a) Amino acid sequence of the zinc-containing DNA-binding domain of the glucocorticoid receptor. There are two zinc atoms within this domain, each bound to four cysteine residues. One of these stabilizes the recognition helix (red) that provides sequence-specific binding to DNA and the other zinc region contains a loop (purple) that is involved in formation of the dimeric molecule. (b) Nucleotide sequence of the region of DNA that binds to the glucocorticoid receptor, the glucocorticoid response element, GRE. The two palindromic half-sites (blue and orange) are separated by a spacer region of three base pairs. (c) The DNA fragment that was used in the crystal structure determination of the complex with the DNA-binding domain of the glucocorticoid receptor.

in Figure 10.9) are not separated into discrete units but are interwoven into a single globular domain with extensive interactions between them. In each of the two zinc motifs the second pair of cysteine zinc ligands initiates an amphipathic α helix. The hydrophobic sides of these two α helices pack against each other to form a compact core with a hydrophobic interior. Hydrophilic side chains on the other side of the first α helix are exposed to solvent and form the DNA interaction area. The two zinc atoms and the

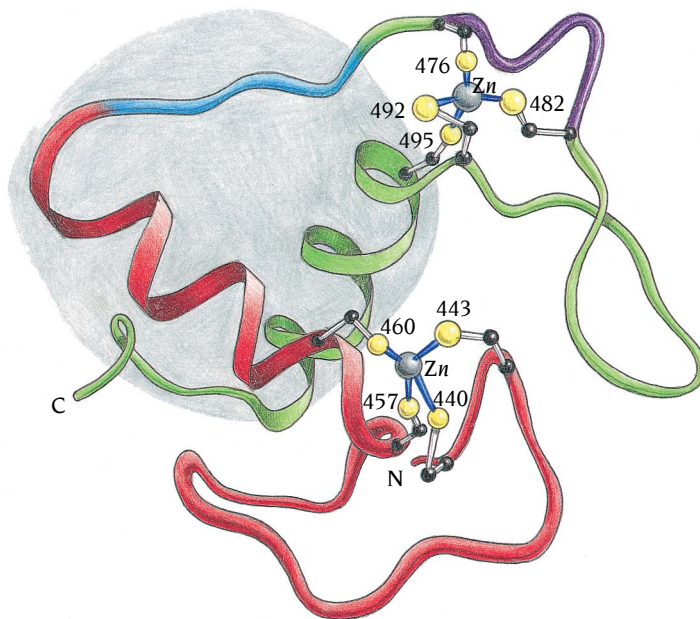


Figure 10.9 Schematic diagram of the three-dimensional structure of the DNA-binding domain of the glucocorticoid receptor. The two zinc-binding regions, defined from the amino acid sequence (see Figure 10.8a) are colored red and green, respectively, and the region that joins them is blue. Each has one α helix and contains a zinc atom bound to four cysteine residues. The two α helices are part of a compact globular core, shown in gray. The dimerization loop is shown in purple. The zinc atoms and the residues between the zinc ligands form protrusions from this globular core. (Adapted from T. Härd et al., *Science* 249: 157–160, 1990.)

protein regions between the zinc ligands form protrusions from this globular core. A comparison of a classic zinc finger (see Figure 10.1b) with the zinc-binding motifs in the glucocorticoid receptor (see Figure 10.9) shows that their three-dimensional structures are quite different. This difference is reflected in their different modes of binding to DNA.

A dimer of the glucocorticoid receptor binds to DNA

The crystal structure of the DNA-binding domain of the glucocorticoid receptor bound to a cognate DNA fragment was determined to 2.9 Å resolution in 1991 by the group of Paul Sigler at Yale University. The regions of DNA that bind to this receptor, the glucocorticoid response elements, GRE, comprise two identical palindromic half-sites, each of six base pairs with the sequence shown in Figure 10.8b. These half-sites are separated by a spacer region of three base pairs whose sequence is essentially unimportant but whose length is crucial for proper binding of the receptor to the two half-sites.

Solution studies have shown that the complete receptor molecule is a dimer that is stabilized by the ligand-binding C-terminal domain. The DNA-binding domains alone in solution are monomers but they dimerize upon binding to DNA. Sigler's group found by serendipity that this dimer interaction is very specific and crucial for binding to response elements with the proper length of the spacer region. They used a DNA fragment (see Figure 10.8c) that contained a spacer of four instead of three base pairs because they wanted an exactly symmetrical DNA fragment. Nevertheless, the monomeric domains dimerized strongly, with the domains arranged such that they could interact correctly only with DNA half-sites separated by the normal three base pairs (Figure 10.10). In the crystal, using a spacer of four base pairs, only one domain interacts properly with one half-site of the DNA. The extra base pair in the spacer displaces the second half-site by one base pair, and as a result the second domain is presented with a noncognate sequence with which it forms a "nonspecific" interface. When presented with a DNA target with the correct three-base pair spacer, both DNA-binding domains bound symmetrically in a specific manner. Aaron Klug's group has since shown that both domains of the estrogen receptor also bind to their cognate DNA half-sites when the spacer has only three base pairs.

The two domains in the dimer bound to DNA are very similar (see Figure 10.10). There are, however, significant differences between the structure of the monomer in solution and that of the subunits in the dimer. Although their overall folds are similar, binding to DNA induces substantial conformational changes especially in the dimer interface area. Most of the interdomain contacts are made by residues in the loop between the first two cysteine zinc ligands of the second zinc region, Cys 476 and Cys 482 (see Figure 10.9). This

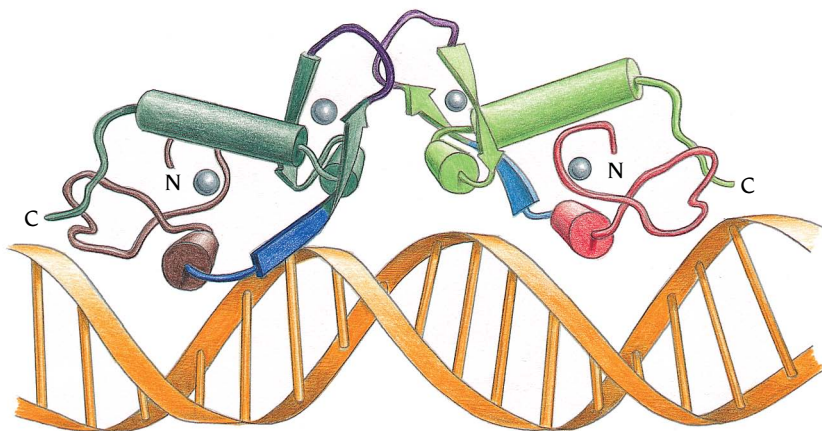


Figure 10.10 Structure of the complex between the dimeric glucocorticoid receptor molecule and a DNA fragment (orange). The two zinc-binding regions of each subunit have different colors; brown and dark green in one subunit and red and light green in the second. The linker region is blue. The recognition helices of the dimer (red and brown) are positioned in the major groove. The distance between them, which corresponds to one turn of the DNA helix, is fixed by the dimerization loop (purple). This region undergoes a significant conformational change when the dimer binds to DNA. (Adapted from B.F. Luisi et al., *Nature* 352: 497–505, 1991.)

loop, called the “D (for dimerization) Box” is not well defined in the monomer structure but is a fixed, well defined β turn in the dimer (see Figure 10.10). Furthermore, the structure around the second zinc is not well defined in the free monomer, but becomes a well defined β -strand-loop-helix in the DNA-bound dimer that forms the intersubunit interface of the dimer. In short, DNA acts as an allosteric effector of its own recognition by supporting tertiary structural changes that lead to a new quaternary structure uniquely suited to bind cooperatively to the correct DNA target.

An α helix in the first zinc motif provides the specific protein–DNA interactions

The two DNA-binding domains in the dimer bind to one face of the DNA, with the interactive surfaces of each subunit in successive major grooves (see Figure 10.10). The amphipathic α helix in the first zinc region, residues 457 to 469 in the glucocorticoid receptor, forms sequence-specific interactions with the edge of the bases in the major groove, and is therefore called the recognition helix. There are no protein contacts to the bases in the intervening minor groove. The two recognition helices of the dimer have the correct orientation and distance between them to fit into two successive major grooves on one side of the DNA double helix. This mode of DNA contact, which is very sensitive to the number of nucleotides separating successive binding sites, bears a striking resemblance to the binding of Cro and repressor to prokaryotic DNA (compare Figures 8.14 and 10.10) but is quite different from the way that two successive classic zinc fingers bind (see Figure 10.3).

The recognition helix is positioned in the major groove by a number of nonspecific contacts between phosphate groups and protein side chains, as in many other protein–DNA interactions. These contacts are made mainly by residues from the two loop regions between the second and third cysteine zinc ligands in both zinc motifs (see Figure 10.8a). One of the nitrogen atoms of His 451 from the first zinc motif in the glucocorticoid receptor forms a hydrogen bond to one of the phosphates of base pair T-A 8. The orientation of the His 451 side chain is defined by a hydrogen bond network at the tip of this loop involving Ser 448. The His 451 residue is conserved in this receptor family, and Ser or Thr always occur at position 448, indicating that this network of hydrogen bonds is of general importance for hormone receptor recognition of response elements. A further important nonspecific interaction is provided by Arg 489 in the second zinc region, which also hydrogen bonds to a phosphate in the DNA backbone. Arg 489 is another conserved residue in this receptor family and its functional importance is shown by a spontaneous mutation that occurs in humans: Arg 489 is replaced with Gln in the vitamin D receptor of patients with hereditary vitamin D-resistant rickets.

Three residues in the recognition helix provide the sequence-specific interactions with DNA

Three residues in the NH_2 terminal half of the glucocorticoid receptor recognition helix—Lys 461, Val 462, and Arg 466—make specific contacts with the edges of the bases in the major groove (Figure 10.11). Arg 466 makes two hydrogen bonds to G4, similar to the Arg–G interaction described previously as a frequent feature of the classic zinc finger–DNA interactions (see Figure 10.6). If Arg is replaced with Lys or Gly, the mutated protein no longer functions. Lys 461 provides one direct and one water-mediated hydrogen bond to G7'. Both Lys 461 and Arg 466 are absolutely conserved in the receptor superfamily, and both their targeted base pairs G-C 4 and C-G 7 occur in most hormone-response elements. Finally, in the glucocorticoid receptor Val 462, which is not conserved within the hormone receptor family, makes a favorable van der Waals contact with the methyl group of T5. Thymine at position

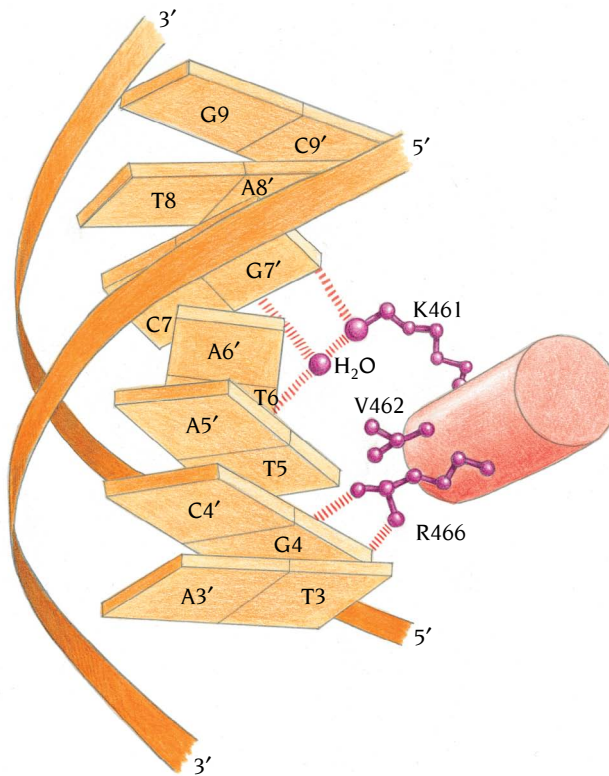


Figure 10.11 Sequence-specific interactions between DNA (orange) and the recognition helix (red) of the glucocorticoid receptor. Three residues, Lys 461, Val 462 and Arg 466 make specific contacts with the edges of the bases in the major groove.

5 is a unique and consistent feature of all glucocorticoid response elements but is not conserved in the response elements of other members of the hormone receptor family.

The two zinc ions fulfill important but different functions in the DNA-binding domains. The first zinc ion is important for DNA-binding because it properly positions the recognition helix; the last two cysteine zinc ligands are part of this helix. The second zinc ion is important for dimerization since the five-residue loop between the first two cysteine zinc ligands is the main component of the dimer interaction area.

The retinoid X receptor forms heterodimers that recognize tandem repeats with variable spacings

Generally, the family of nuclear receptors have low target affinity, high amino acid sequence homology and similar DNA response elements. How then are they targeted to their appropriate genes? The steroid hormone receptors bind exclusively as homodimers to response elements where the half-sites are organized in a palindromic orientation. Other nuclear receptors, such as those for vitamin D (VDR), thyroid hormone (TR) or *trans*-retinoic acid (RAR), form heterodimers with the *cis*-retinoic acid receptor (RXR) that bind to response elements with half-sites organized as direct repeats. Different heterodimers recognize response elements with different lengths of the spacer region. Paul Sigler's group has determined the x-ray structure of the heterodimer of the DNA-binding domains of RXR and TR bound to DNA, and has suggested a mechanism by which different heterodimers involving the common partner RXR can recognize response elements of identical target sequences but with different lengths of the spacer regions.

The individual domains of the two receptors both have structures similar to that of the glucocorticoid receptor, and they bind to DNA in a similar way, with their recognition helices in the major groove. The dimer contacts are, however, totally different. In the glucocorticoid receptor, which binds to a palindromic DNA sequence like the 434 repressor described in Chapter 8, the domains interact symmetrically in a head-to-head fashion: equivalent

residues in the two domains interact across a twofold symmetry axis. By contrast, in the RXR-TR heterodimer, which binds to direct DNA repeats like the classic zinc finger transcription factors, the domains bind to each other in a head-to-tail fashion: residues in one domain interact with totally different residues in the other domain. As a consequence the dimer is polar, as is the response element, and in the complex the RXR domain binds exclusively to the 5' half-site of the response element even though the two direct repeats have the same nucleotide sequence.

The two domains are oriented in the dimer in such a way that each recognition helix binds to its half-site in the major groove of B-DNA with a spacer of four base pairs between the half-sites. The dimer interface region straddles the minor groove of the spacer region. Residues from the second zinc motif of RXR interact with residues in the first zinc motif of TR. These residues are also involved in nonsequence-specific interactions with the DNA backbone. Thus the DNA contacts support the dimer interface, which in turn supports both the specific and nonspecific contacts with DNA. These mutually supporting interactions between the protein domains and the DNA backbone reinforce a lesson learned from the glucocorticoid receptor-DNA complex concerning the importance of spacing between the half-sites: the DNA-binding domains of the nuclear receptors, which are monomers in the absence of DNA and which generally bind weakly as monomers to a single half-site, bind with high affinity and strong cooperativity to response elements in which the appropriate half-sites are spaced correctly.

RXR forms heterodimers with VDR, TR and RAR that bind to response elements that differ only in their inter half-site spacing (Figure 10.12), of three, four and five base pairs, respectively. The DNA-binding domains of these four nuclear receptors have highly similar amino acid sequences and hence similar three-dimensional structures. How can RXR form dimers with these similar partners such that their two recognition helices can bind to the corresponding two specific B-DNA target sequences in the major groove whose relative positions are different both in orientation and length of separation due to the different lengths of the spacer between them? Using the structure of the RXR-TR-DNA complex, Sigler's group has tried to answer this question by modeling possible structures for the other two dimers, assuming that RXR and its binding mode is the same in these complexes. They found that by rotating the second RXR domain about 35° together with a translation of 3.4 Å, plausible new interface regions occur that allow dimers to form with the appropriate DNA-binding properties for the different complexes. It is significant that the amino acid residues that participate in these interface regions occupy nonconserved positions within the receptor family and are relatively specific to a particular receptor, thereby ensuring a unique mode of dimerization on the appropriate response element.

In summary, a DNA-supported asymmetric interface located within the DNA-binding domains of these nuclear receptors provides the molecular basis for receptor heterodimers to distinguish between closely related response elements. RXR can provide a repertoire of different dimerization surfaces, each one unique for a specific partner, allowing dimers to form that are adapted to the length of the spacer region in their corresponding response elements.

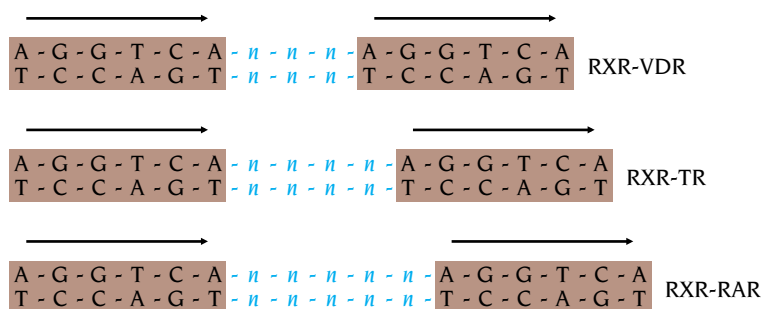


Figure 10.12 Response elements for heterodimers of the nuclear receptor for *cis*-retinoic acid (RXR) with the receptors for vitamin D (VDR), thyroid hormone (TR) and *trans*-retinoic acid (RAR). The half-sites of these response elements have identical nucleotide sequences and are organized as direct repeats. They differ in the number of base pairs in the spacer region between the half-sites. This difference forms the basis for the ability of the heterodimers to discriminate between the different response elements.

Yeast transcription factor GAL4 contains a binuclear zinc cluster in its DNA-binding domain

A third subfamily of zinc-containing motifs has been found in several transcription factors from yeast and fungi. The most thoroughly studied member of this family, GAL4, activates transcription of genes required for the breakdown of the sugars galactose and melibiose. GAL4 binds as a dimer to DNA recognition sequences that are 17 base pairs in length, and some of the genes that are activated by GAL4 have four of these upstream-activating sequences (UAS).

The GAL4 protein is large, having 881 amino acids, and different functions have been assigned to several regions of the polypeptide chain. In addition to the DNA-binding region, residues 1–49, and three acidic activating regions, residues 94–106, 148–196 and 768–881, there is a dimerization region formed by residues 50–94. The structure of a GAL4 fragment comprising residues 1–65 complexed with a DNA sequence of 19 base pairs has been determined by the group of Stephen Harrison, Harvard University. This protein fragment, which binds as a dimer to DNA, can be divided into three structurally and functionally distinct regions: a zinc cluster domain at the N-terminus, a dimerization region at the C-terminus and a 9-residue linker region in between (Figure 10.13).

Residues 50–64 of the GAL4 fragment fold into an amphipathic α helix and the dimer interface is formed by the packing of these helices into a coiled coil, like those found in fibrous proteins (Chapters 3 and 14) and also in the leucine zipper families of transcription factors to be described later. The fragment of GAL4 comprising only residues 1–65 does not dimerize in the absence of DNA, but the intact GAL4 molecule does, because in the complete molecule residues between 65 and 100 also contribute to dimer interactions.

In contrast to the zinc-containing DNA-binding domains described so far, the GAL4 family contains a cluster of two zinc atoms liganded to six cysteine residues, and two of these cysteines (residues 11 and 28) are bound to

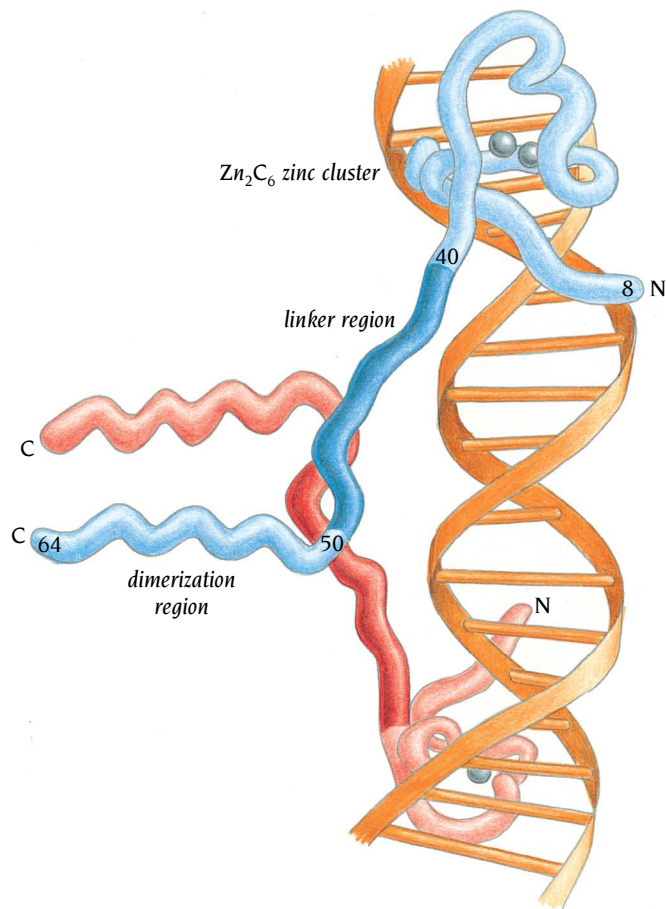


Figure 10.13 Schematic diagram of the dimeric GAL4 transcription factor bound to DNA. The structure of each subunit is divided into three distinct regions: a dimerization region joined by a linker region to a DNA-binding Zn_2Cys_6 zinc cluster region. The two dimerization α helices are packed into a coiled coil like a leucine zipper, and the length of the linker regions determines the distance between the DNA-binding sites. (Adapted from R. Marmorstein et al., *Nature* 356: 408–414, 1992.)

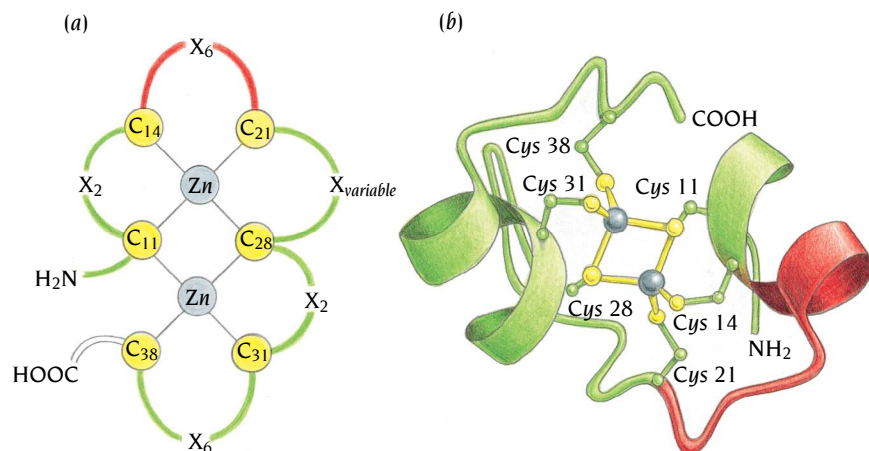


Figure 10.14 Schematic diagrams of the DNA-binding zinc cluster region of the GAL4 subunit. (a) The zinc cluster contains two zinc atoms each bound to four cysteine residues, two of which bridge the zinc atoms. The diagram illustrates the number of amino acids in the loop regions between the cysteine ligands. (b) Richardson-type diagram of the DNA-binding region. The red region provides the sequence-specific DNA interactions. The zinc cluster stabilizes the structure to give the proper fold for DNA binding.

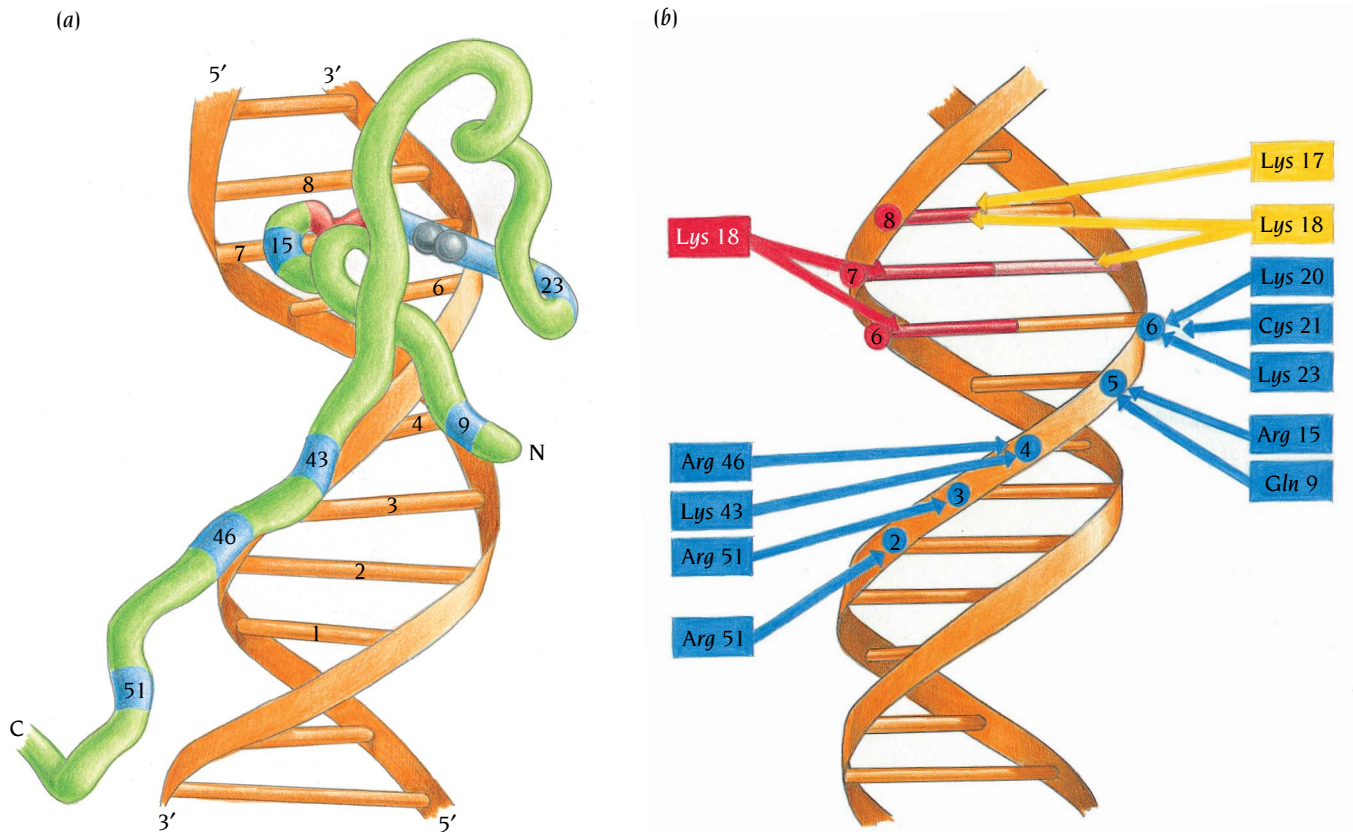
both zinc atoms, forming two bridges between them (Figure 10.14). The amino terminal polypeptide chain of GAL4 wraps around the zinc cluster and contains two short α helices, one of which provides the specific interactions with DNA (red in Figure 10.14b). It is clear from the structure that the function of the zinc cluster is to ensure the proper fold for DNA-binding by this region of the polypeptide chain.

The zinc cluster regions of GAL4 bind at the two ends of the enhancer element

The 19-base pair DNA fragment used in the structure determination contains a GAL4 upstream-activating sequence (UAS) of 17 base pairs that is semi-palindromic, with a highly conserved CCG sequence at either end, reading outwards 5' to 3' from the center of the sequence. These two conserved, three-base pair recognition modules are always separated by 11 base pairs in UAS that bind GAL4. In the UAS of each member of this subfamily of enhancer elements the length of the spacer between the two CCG sequences is constant and characteristic, but other subfamilies have spacers of different lengths.

The zinc cluster modules of the GAL4 dimer bind in the major groove of DNA, in contact with the CCG base pairs at the extreme ends of the consensus region (see Figure 10.13). Since the DNA in the complex is essentially B-DNA, these binding sites are separated by more than one full turn of DNA. The two zinc cluster modules bind in an almost identical manner to the two sites, and they are anchored to the sugar-phosphate backbone by hydrogen bonds to phosphates 5 and 6 (Figure 10.15). These interactions position the module so that the C-terminus of the first α helix (red in Figure 10.14b) points into the groove. At the C-terminus of an α helix the C=O groups of the main chain are not involved in helical hydrogen bonds (see Figure 2.2) but point away from the helix. Consequently in the GAL4–DNA complex these main-chain carbonyl groups are directed towards the edge of the bases in the major groove. The carbonyl group of residue 17 accepts a hydrogen bond from cytosine 8' and the carbonyl of residue 18 accepts hydrogen bonds from both cytosines 7 and 8'. In addition, the side chain of Lys 18 donates hydrogen bonds to both guanines 6' and 7'. This pattern of contacts uniquely specifies the CCG sequence at base pairs 6 to 8 (see Figure 10.15). There are no other direct interactions between the protein and DNA base pairs.

The GAL4 recognition module therefore contains only one protein side chain, Lys 18, that provides specific interactions with the DNA. The remaining specific interactions with DNA are from main-chain atoms and depend critically on the correct conformation of the protein. The correct positioning of the C-terminus of the α helix is particularly important for recognition. This is to date the only example of a protein–DNA interaction in which



the main-chain carbonyl groups at the C-terminus of an α helix participate in specific DNA contacts. Since Lys 18 is conserved in most members of this family of transcription factors, it is quite clear that all their recognition modules bind in the same way to DNA, and that discrimination between different upstream-activating sequences depends on factors other than the zinc-containing modules.

The linker region also contributes to DNA binding

The linker region connecting the recognition zinc module with the dimerization region of the GAL4 fragment is an extended chain (see Figure 10.13). There are several nonspecific contacts from this linker region and the beginning of the dimerization region to phosphate groups of the DNA (see Figure 10.15). The path of each linker follows one strand of the DNA backbone from the zinc module towards the minor groove, thus positioning the dimerization domain over the minor groove. In addition, the structure formed by residues 47–51 of both subunits is sterically complementary to the phosphate backbones of the minor groove, and this contributes to the binding affinity. It is obvious from the structure of this GAL4 fragment that the linker regions define the distance between the two sites at which the recognition modules of the dimer bind to the DNA. A large region of the DNA major groove, between the binding sites at the ends of the 17-base pair DNA fragment, is not covered by the protein and is accessible to solvent (see Figure 10.13).

In conclusion, recognition of DNA enhancer elements by a GAL4 dimer involves three components: (1) a specificity for the conserved CCG triplets, which is provided by direct contacts between the major groove bases and the C-terminus of an α helix in the zinc-containing recognition modules; (2) a requirement for symmetrical sites, provided by a coiled-coil dimerization region that lies over the minor groove; and (3) a preference for a spacer between the CCG triplets of 11 base pairs, which is imposed by the structure of the linker region.

Figure 10.15 Schematic diagrams of the binding of one subunit of GAL4 to DNA. (a) Shows the structure of the DNA-binding domain in complex with the DNA; and (b) shows the interactions between amino acid residues and the DNA. The zinc cluster region binds in the major groove of DNA and is anchored to the sugar-phosphate backbone by nonsequence-specific interactions (blue). The C-terminus of the first α helix points into the major groove, and two main-chain C=O groups from residues 17 and 18 (yellow) form hydrogen bonds to the edge of the bases. There is only one sequence-specific interaction with an amino acid side chain, which is provided by Lys 18 (red). The linker region is an extended chain that follows one strand of the DNA and provides several nonspecific contacts (blue) to the DNA. The numbering of the base pairs starts from the center of the DNA fragment. (Adapted from R. Marmorstein et al., *Nature* 356: 408–414, 1992.)

DNA-binding site specificity among the C₆-zinc cluster family of transcription factors is achieved by the linker regions

At least 11 other fungal DNA-binding proteins are known to contain repeated Cys sequences like those found in the GAL4 recognition module. Residues 15–20 of GAL4, which are in closest proximity to DNA in the complex, are highly conserved in these homologous proteins. Lys 18, which makes the only direct sequence-specific contact with DNA, is conserved in all but three cases. In addition, these proteins all recognize CCG triplets in their upstream-activating sequences. Presumably, therefore, all the individual zinc-containing DNA-binding motifs of these proteins bind to CCG triplets in the same way. In which case how do the proteins recognize their proper upstream-activating sequences?

The group of Mark Ptashne at Harvard University, has tried to answer this question by domain-swapping experiments between GAL4 and the related yeast transcription factor, PPR1. Both Lys 18 and Lys 19 in GAL4 are conserved in PPR1. Both proteins recognize a DNA site containing two palindromic CCG triplets, but the triplets are separated by 6 base pairs in the enhancer elements controlled by PPR1 instead of the 11 in those regulated by GAL4. The PPR1 protein binds *in vitro* only to DNA fragments with a spacer of 6 base pairs and GAL4 only to those with 11 base pairs.

The Ptashne group first constructed a chimeric protein comprising the DNA-binding zinc module of PPR1 (residues 6–40 in the GAL4 numbering system) and the linker and dimerization regions of GAL4 (residues 41–100) (Figure 10.16). In spite of having the DNA-binding module of PPR1, this protein had the binding specificity of GAL4; in other words, it bound much more strongly to DNA fragments in which the CCG triplets were separated by 11 base pairs than by 6 base pairs. To determine if the linker or the dimerization region is responsible for this specificity, they constructed a second chimeric protein containing the zinc module, the linker region and the first residues of the dimerization module (residues 6–57) from PPR1 and the main part of the dimerization module (residues 58–100) from GAL4. In contrast to the first chimera, this protein has a binding specificity for PPR1 binding sites that is 200 times stronger than for GAL4 sites. Similar results have been obtained using a third member of this family which has a spacer of 10 base pairs between the CCG triplets.

Subsequently Stephen Harrison's group determined the x-ray structure of a PPR1–DNA complex and showed that the zinc cluster domain of PPR1 and its mode of binding to DNA was very similar to that of GAL4, and that PPR1 also dimerized through a coiled-coil region. However, the linker region

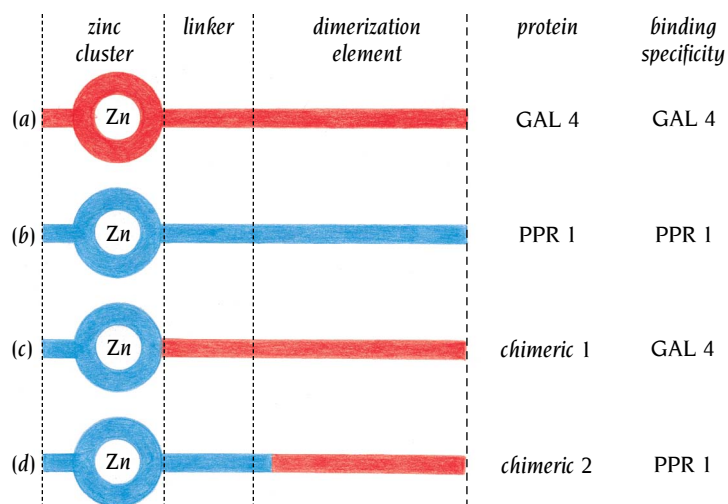


Figure 10.16 Domain-swapping experiments with GAL4 (a) and a related transcription factor PPR1 (b) have shown that the specificity of DNA binding depends on the size of the linker region. A chimeric protein with the PPR1 zinc cluster combined with GAL4 linker and dimerization regions (c) has GAL4 DNA-binding specificity, whereas a chimeric protein with PPR1 zinc cluster and linker regions combined with a GAL4 dimerization region (d) has PPR1 specificity.

was very different, comprising a β hairpin instead of an extended chain. This gives a more compact asymmetric dimer, and since the DNA-binding zinc cluster regions are closer together they are adapted to bind a DNA region with a spacer of six base pairs between the two CCG recognition sequences.

These results prove that the zinc cluster domains of this family of transcription factors work interchangeably to recognize the conserved CCG triplets, while the region of about 20 amino acids comprising the linker and the beginning of the dimerization domain directs the protein to its proper upstream-activating sequences, with its appropriately sized spacer. Zinc and its ligands fix the protein chain in a conformation that allows specific DNA binding.

Families of zinc-containing transcription factors bind to DNA in several different ways

We have now seen examples of three different ways in which zinc-containing transcription factors recognize their DNA-binding sites, and how the specificity of the individual members within these three families is achieved. In all three cases the DNA recognition module consists of a small polypeptide chain wrapped around a zinc atom, such that an α helix is exposed to bind into the major groove of DNA. But this is achieved in three different ways with three different conformations of the polypeptide chains. The common role of zinc and its ligands is to fix firmly the conformation of the polypeptide chain, so that the crucial α helix is held in a position to make specific contacts to exposed bases of DNA, while the rest of the polypeptide chain anchors the recognition module to the major groove of DNA by nonspecific contacts to the DNA backbone.

The classic zinc fingers bind in tandem to successive recognition sites in the major groove of DNA. Variability within the family is achieved by varying the number of zinc fingers as well as the side chains of their recognition helices. Members of the nuclear receptor family bind to DNA response elements with very similar target sequences. Specificity is achieved in several different ways; the target sequences are organized as palindromes or direct repeats with linker regions of different lengths, and the DNA-binding domains can form homo- or heterodimers. Finally, the GAL4 family of transcription factors all recognize conserved CCG triplets at the ends of the enhancer elements in the same way, using interchangeable zinc cluster recognition modules. The specificity of recognition is achieved by varying the linker regions between the dimerization domains and the recognition modules, so that they are compatible with the different lengths of the spacer DNA between the CCG triplets in the different enhancer elements.

The world of zinc-containing DNA-binding proteins is by no means exhausted by these three subfamilies. Several other subfamilies are already known with different three-dimensional structures and different sequence patterns of cysteine and histidine residues that form the zinc ligands. Further subfamilies may well be discovered as the genomes of different species are sequenced; whether or not any fundamentally new principles for DNA-protein recognition will be discovered amongst these new subfamilies remains to be seen.

Leucine zippers provide dimerization interactions for some eucaryotic transcription factors

The leucine zipper motif (see Chapter 3) was first recognized in the amino acid sequences of a yeast transcription factor GCN4, the mammalian transcription factor C/EBP, and three oncogene products, Fos, Jun and Myc, which also act as transcription factors. When the sequences of these proteins are plotted on a helical wheel, a remarkable pattern of leucine residues